



Chips: The biggest AI bottleneck

May 11, 2026

- This week's upcoming Xi-Trump summit throws the geopolitical issues surrounding semiconductors back into the spotlight.
- In this paper, we argue that the key bottlenecks are in advanced chip and, to a lesser extent, specialized equipment (e.g., EUV lithography tools) and manufacturing, located predominantly in Taiwan and the Netherlands.
- Technology, particularly advanced chips, is now seen as the heart of an intense competition to retool economies, reform militaries, and serve as a source of national strength.

Our new series, "The Great Rebalancing," focuses on five "critical chokepoints" that we see as redefining global power: **Currencies, Critical Minerals, Chips, Cables, and Cloud.**

Each chapter of our series will explore **how and why these chokepoints define a new matrix of global power.** Currencies allocate risk, critical minerals anchor the physical economy, chips generate compute power, cables move data and finance, and cloud platforms organize markets. Influence over this layered system shapes geopolitical standing – determining which nations can scale their economies, exert control, and apply strategic pressure.

This series argues that a "great rebalancing" is underway, shifting the global order from liberal institutionalism, where interdependence fosters cooperation, toward what we term "infrastructure realism". In this new paradigm, control over key systems and inputs shapes outcomes. Under "infrastructure realism", material and digital infrastructures become decisive instruments of state power. The US-China trade tensions of 2025 starkly illustrated this shift. The financial system — from bond markets to payment rails to digital money — acts as the meta-infrastructure that coordinates and prices all others.

Thus, this series addresses the wider question: which major powers are best positioned for influence and what are their strategic advantages? The 2025 US National Intelligence Annual Threat Assessment identified China as the primary actor challenging US global interests, citing its use of "asymmetric and conventional hard power tactics" and its drive to establish "alternative systems" to compete with the US¹. While the competitive balance of power is longstanding, 2025 highlighted the asymmetric economic dependencies and capabilities shaping the relationship between the two countries.

As these five chapters will demonstrate, we believe power now rests on the control of critical infrastructure. The US and China are in a race to reduce strategic dependencies on each other, and 'we believe the outcome of this dynamic' will define the next era.

Authors

Marion Laboure
Research Analyst

Camilla Siazon
Research Analyst

¹ Review of ANNUAL THREAT ASSESSMENT of the U.S. INTELLIGENCE COMMUNITY. 2025. Office of the Director of National Intelligence.



Introduction: From compute to chokepoints, the competition for industrial control

In Chapter 1: Currency, we argued that the US is successfully adapting to maintain leadership in digital finance (or, as we call it, “[meta-infrastructure](#)”), whilst in Chapter 2: Critical Minerals, we [analysed China’s dominance](#) across different segments of the critical minerals supply chains and how the central government has been able to leverage this advantage in recent national security disputes. These case studies illustrate each layer of the infrastructure-realist stack, which we describe as digital, invisible, and embedded within complex, globally distributed supply chains.

In this paper, we now move on to the next layer: semiconductors, otherwise colloquially referred to as chips. As the contours of modern conflict shape themselves around the AI race, chips have become central to great power competition given their crucial bottleneck in the AI stack.

We examine the cutting-edge chip technologies that underpin cloud computing and modern defence systems, with chipmakers emerging as some of the most visible beneficiaries of the AI-driven technological cycle. We also examine the US-China conflict around legacy chips, the mature nodes (>22nm) that power everyday appliances.

Because both the US and China have used their respective industrial strengths in chips and critical minerals for retaliatory actions, and because chips themselves are a direct end-product of critical minerals, it is tempting to draw neat parallels between the two countries’ economic toolkits. However, there are key differences to consider. First, the scale of investment and engineering complexity in semiconductors is far greater than that of rare earths. Second, unlike critical mineral supply chains – where China unquestionably exerts dominance across multiple stages – the semiconductor ecosystem is more fragmented, with different countries controlling distinct parts of the value chain. Apple’s complex production process illustrates this asymmetry: chips are designed by Apple in the US, manufactured in Taiwan, and assembled in China².

At the centre of this competition lies Taiwan. US Treasury Secretary Scott Bessent summed up the Taiwan risk at Davos earlier this year: “The single biggest threat to the world economy” he said, “is that 97% of the world’s most advanced chips are manufactured on the island. If that island were blockaded, if that capacity were destroyed, it would be an economic apocalypse³.” An equally critical though less immediate bottleneck is the Dutch company ASML; without its unique lithography equipment, Taiwan cannot build the fabs that produce these integrated circuits. However, ASML has indicated a potential to scale up production by 3x by 2030, if necessary.

Ultimately, building a vertically integrated domestic semiconductor industry has so far proven to be an unrealistic goal for any single nation. As we illustrate, each stage of the chip supply chain is a potential chokepoint for another. This creates a competitive landscape with no clear overarching winners, resulting in structural interdependence, supply fragility, and a stalemate that keeps both the US and China in an asymmetric power struggle.

² For a thorough breakdown of Apple’s supply chain, read Chris Miller and Vishnu Venugopalan’s work here: Chris Miller and Vishnu Venugopalan, “Apple’s Supply Chain: Economic and Geopolitical Implications,” American Enterprise Institute, June 3, 2025, <https://www.aei.org/research-products/report/apples-supply-chain-economic-and-geopolitical-implications/>.

³ Tripp Mickle, “The Looming Taiwan Chip Disaster That Silicon Valley Has Long Ignored,” *The New York Times*, February 24, 2026, <https://www.nytimes.com/2026/02/24/technology/taiwan-china-chips-silicon-valley-tsmc.html>.



1. Chips: Infrastructure as statecraft

Although China dominates in mineral processing and magnets manufacturing, the US maintains advantage in their downstream outputs: chip software, intellectual property (IP), and manufacturing equipment. As a result, the US has been able to counter China's critical minerals leverage by deploying its own toolkit of semiconductor dominance. This dynamic was starkly illustrated in April 2025, when the US imposed export controls on Nvidia's H20 chips to China in response to Beijing's own rare earth export restrictions on April 4⁴.

The ongoing chip competition illustrates how crucial semiconductors have become in contemporary economic statecraft. Here, we trace their historical emergence as strategic assets, analyze their diverse economic functions and market structures, and look at how the race for supply chain sufficiency has narrowed to a contest over cutting-edge chip technology.

1.1 Semiconductors: The physical layer of the digital economy

Semiconductors – miniature electronic devices primarily made from silicon or germanium – are indispensable components of nearly every modern electronic device. They power everything from smartphones, computers and home appliances to critical infrastructure such as telecommunications networks, data centers, autonomous vehicles, and industrial automation systems. At the highest level, semiconductors are designed to sense, compute, store, transmit, and control information and energy.

The overall semiconductor market is expected to generate ~\$1trn in revenue in 2026, and over \$1.5trn in 2030, compared to ~\$791bn in 2025⁵. Today's semiconductor industry is highly concentrated, with a few dozen major players producing a wide range of specialized chips with distinct economic functions:

- **Logic chips (“Compute”)**: These act as the “brains” of digital systems, processing information and executing tasks. This category includes CPUs (central processing units), GPUs (graphics processing units), and AI accelerators used in data centers, personal computers, smartphones, and cloud infrastructure.
- **Memory chips (“Storage”)**: These chips store and retrieve data, enabling systems to retain information and operate efficiently. Key examples include DRAM and NAND flash, which are ubiquitous across servers, PCs, and mobile devices⁶.
- **Analog and power chips (“Control”)**: These manage electrical signals and power flows, allowing digital systems to interact with the physical world. They are essential in automotive systems, industrial machinery, energy infrastructure, and defense applications.
- **Specialized and integrated chips (“Optimized Systems”)**: These are designed for specific tasks or highly integrated functions. This includes application-specific integrated circuits (ASICs) tailored for particular workloads (e.g. crypto mining or networking) and systems-on-a-chip (SoCs) that combine multiple components into a single, power-efficient unit.

⁴ Marion Laboure and Camilla Siazon, “Thematic Research: US vs. China: The Biggest Bargaining Points”, Deutsche Bank Research, April 17, 2025, https://research.db.com/research/Article?rid=e0c6c6ad-47e1-41cb-8162-edf715f0355a-604&kid=RP0001&documentType=R&wt_cc1=IND-3146993-0000.

⁵ Dylan Peterson, “Global Annual Semiconductor Sales Increase 25.6% to \$791.7 Billion in 2025 - Semiconductor Industry Association,” Semiconductor Industry Association, February 6, 2026. <https://www.semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025/>. For the 2030 projections, see Company, Manufacturing. “2026 Technology Symposium Highlights.” Tsmc.com. Taiwan Semiconductor Manufacturing Company. https://www.tsmc.com/english/symposium_highlights/2026.

⁶ DRAM and NAND are types of memory semi-components, but DRAM memory manages data and requires power, therefore making it volatile, whereas NAND stores data and does not require power.



1.2 A brief history of semiconductors: From Cold War origins to global bedrock

Since their invention in the 1940s, semiconductors have been intrinsically linked to national security and strategic competition. Their role, however, has evolved from state-led innovation to market-driven globalization, and more recently back to geopolitical rivalry. Semiconductors were also once in the wheelhouse of US strength, but over the 1990s-2010s, the country's economic model was guided by the belief that integrating global trade and supply chains promoted peace and economic prosperity. As we note later, the symbiotic relationship that flourished between Taiwan's foundries and US design firms exemplified this era.

State-led origins: Semiconductors as military technology (1940s–1980s)

Semiconductors emerged from Cold War US defence programs, most notably with the invention of the transistor (a single, fundamental chip component used to switch or amplify electrical signals) at Bell Labs (1947-58). Early innovation was heavily subsidized and driven by military and space exploration demand.⁷

In the late 1950s, the invention of the integrated circuit (a chip that contains millions or billions of interconnected transistors) helped semiconductors move from niche military applications into the rapidly expanding commercial computer industry, fuelling the rise of Silicon Valley. By the 1980s, semiconductors were embedded across everyday technologies like cars and personal computers, as exponential improvements in computing power and cost efficiency led to a rigorous and sustained cycle of innovation.

During the initial years of chip innovation, the US remained the relatively unchallenged global leader in semiconductor production, accounting for 60% of supply in the 1970s.⁸ However, as global interest in chip production grew and other countries began to develop their own capabilities, Japan's rapid ascent in the global DRAM market stirred concern among US policymakers, who began to view the country as a primary strategic competitor. By the mid-1980s, Japan had overtaken the US in semiconductor production, and towards the end of the decade, Japanese firms held leading positions in key segments of the semiconductor equipment market.⁹

In response to what it viewed as a national security threat, the US forced the issue. The resulting 1986 US-Japan Semiconductor Agreement sought to halt the dumping of Japanese DRAMs in third markets at below-cost prices and increased foreign access to Japan's market, with Japan agreeing to help secure 20% of its domestic semiconductor market for foreign producers within five years. Concurrently, the US moved to strengthen its own industry by establishing SEMATECH in 1987, a government-backed research consortium comprising of 14 different domestic chip companies.¹⁰

This intense rivalry proved short-lived. Japan's chip ascent stalled following its financial crisis in the early 1990s, and by the end of the 20th century, other US technology competitors like the Soviet Union and South Korea had also faded.

Market-led globalization: Fragmentation and the rise of Taiwan (1990s–2010)

With the dissolution of the USSR in 1991, the US was at its apogee as the standard-bearer for a Western model of liberal democracy and capitalism. This unipolar moment ushered in an era defined by liberal institutionalism, where successive US administrations promoted open markets, free trade, and neoliberal policies through multilateral efforts.

⁷ For more background on the history of semiconductors, see Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (New York: Simon and Schuster, 2022).

⁸ Emily G. Blevins, Michaela D. Platzer, Karen M. Sutter, "Semiconductors: U.S. Industry, Global Competition, and Federal Policy," CRS Report R46581 (Washington, DC: Congressional Research Service, October 26, 2025), <https://www.congress.gov/crs-product/R46581>.

⁹ Japan was supplying 70% of the world's lithography equipment. See Chris Miller's *Chip War*.

¹⁰ Government funding for SEMATECH stopped in the 1990s, after which it relied on industry funding. In 2015, the research group was absorbed by the State University of New York Polytechnic Institute. See Blevins, Platzer, and Sutter, "Semiconductors."



Under this new world order, the semiconductor industry became increasingly globalized and specialized. Its value chain fragmented into distinct segments—design, manufacturing, and equipment—each optimized for efficiency and scale. Landmark trade agreements, including the World Trade Organization’s Information Technology Agreement in 1996 and its 2015 expansion, helped bring semiconductor tariffs close to zero in many countries.¹¹

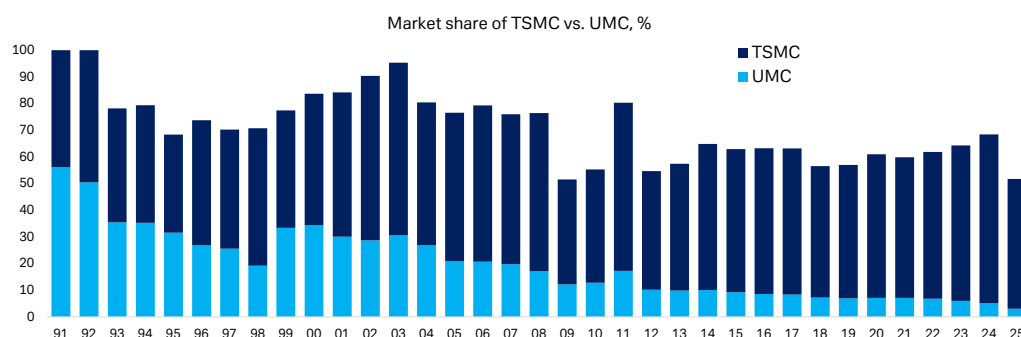
Within this liberalized framework, the industry’s centre of gravity shifted toward Taiwan. Its rise was spearheaded by the Taiwan Semiconductor Manufacturing Company (TSMC), founded in 1987 by Morris Chang, a veteran of US company Texas Instruments. TSMC pioneered the foundry model – manufacturing chips designed by external customers – which in turn enabled the rise of “fabless” design firms, particularly in the US. Prior to this innovation, most chip companies had operated as integrated device manufacturers, a capital-intensive model that required them to design and produce their circuits in-house.

Rather than a strategic rivalry, this division of labour created a powerful symbiotic relationship: the US led in design, while Taiwan specialized in manufacturing. Supported by talent flows from Silicon Valley, this model underpinned two decades of rapid innovation and global integration. TSMC’s rise as a leading foundry enabled the growth of US companies in turn; and this symbiotic relationship sustained both sides for the next two decades. While the increasing reliance on a “fabless” business model would eventually be viewed as a strategic vulnerability for US manufacturing, this trade-off was accepted by US leaders during the post-Cold War era of globalization and liberal institutionalism.

Return of geopolitics: Semiconductors as strategic infrastructure (2010–present)

By 2010, the growing importance of semiconductors in cloud computing, and advanced military systems brought their strategic importance back to the forefront. Chips were no longer viewed merely as commercial inputs, but as foundational infrastructure underpinning economic and military power. This shift marked the beginning of a new phase of geopolitical competition, particularly between the US and China. Consequently, governments have re-engaged with the sector through industrial policy, export controls, and national security frameworks, transforming the industry from a globalized, efficiency-driven system to one increasingly defined by strategic competition and the race to control critical supply chain chokepoints.

Figure 1: TSMC quickly surpassed competitors in the foundry market after its founding, largely due to continued investment even during downturns and a strong drive to innovate beyond companies like UMC.



Source: Deutsche Bank Research, Bloomberg Finance LP.

¹¹ See Blevins, Platzer, and Sutter, “Semiconductors.”



1.3 China's semiconductors policy: From integration to strategic self-reliance

Today, Huawei is a central player in China's drive for both chip design and manufacturing, largely in response to US export controls. Through its chip design arm, HiSilicon, the company has played a key role in developing advanced semiconductors domestically. For much of its history, however, Huawei operated primarily in telecommunications infrastructure, a sector in which it eventually grew into a global giant, helping position China as a leader in 5G and electronics. This success, however, came at a cost: its expanding global footprint turned it into a focal point of geopolitical tensions, as several countries began to assert that Huawei posed national security risks.

The rise of Huawei and China's broader electronics ecosystem would not have been possible without sustained access to foreign semiconductors. Here, we trace China's transition from a mass importer of semiconductors towards its own path of chip independence.

Setting up the scene: China's chip origins (1980s-2000s)

China has been the largest single market for semiconductor exports since the country underwent its technological revolution in the 1980s. Under Deng Xiaoping's program to modernize the economy via science and technology, China transitioned from a closed, centrally planned system toward a market-oriented economy, laying the groundwork for a domestic technology sector. From this period of reform emerged firms such as Huawei, which was founded in 1987 to supply China's growing demand for telecommunications equipment.

Throughout the 1990s and 2000s, China relied heavily on imports from the US, Japan, and increasingly Taiwan to power its industrial expansion. As China entered the 2000s, the government began to make early moves toward domestic semiconductor manufacturing, most notably through the establishment of Semiconductor Manufacturing International Corporation (SMIC) in 2000. Progress, however, remained limited.

Early moves: China's move towards self-reliance amidst heightening geopolitical tensions (2010s-2020s)

By the 2010s, China accounted for ~60% of global semiconductor demand but only ~13% of global supply, underscoring a deep structural dependence on foreign technology¹². This gap became a central vulnerability in China's economic model.

Recognizing this vulnerability, the government began to scale up state support. In 2014, Xi Jinping launched the National Integrated Circuit Industry Investment Fund (the "Big Fund") to finance domestic foundries, design firms, and equipment suppliers. With support from both central and local governments, capital flowed into fabrication plants and upstream supply chains, enabling firms like SMIC to reach 14nm production by 2019¹³.

China's ambition to become a global leader in high-tech manufacturing was further formalized through its "Made in China 2025" strategy. This industrial policy, unveiled in 2015, sought to reduce reliance on foreign technology and help the nation avoid the "middle-income development trap", setting an aggressive target of 70% semiconductor self-sufficiency by 2025.

Unlike the earlier semiconductor success stories of Japan, South Korea and Taiwan – which were deeply integrated into the US-led ecosystem – China pursued, and continues to pursue, a more state-driven and increasingly independent approach aimed at building domestic capabilities across the full semiconductor value chain. Thus, China's scope and

¹² James McBride and Andrew Chatzky, "Is 'Made in China 2025' a Threat to Global Trade?" Council on Foreign Relations, August 2, 2018, <https://www.cfr.org/backgrounders/made-china-2025-threat-global-trade>.

¹³ Anton Shilov, "SMIC Mass Produces 14nm Nodes, Advances to 5nm, 7nm," *Tom's Hardware*, September 16, 2022, <https://www.tomshardware.com/news/smic-mass-produces-14nm-nodes-advances-to-5nm-7nm>.



scale prompted concerns among US officials, with some members of Congress describing Beijing's methods for independence as ones that "violate, circumvent, or otherwise challenge current global trade rules."¹⁴

During this period, several geopolitical episodes also exposed the risks of China's reliance on foreign semiconductor technology, contributing to a broader deterioration in bilateral relations and serving as a wakeup call for the government to accelerate its push for technological self-reliance.

The first was the 2013 Snowden revelations, which highlighted the extent of US surveillance capabilities and amplified Chinese concerns over dependence on foreign digital infrastructure¹⁵. Then in 2018, the US imposed export restrictions on Zhongxing Telecommunication Equipment (ZTE) Corporation, a major Chinese telecommunications equipment and network systems company, for violating US sanctions by supplying equipment to Iran and North Korea. During this period, US semiconductor firms like Qualcomm and Intel were prohibited from supplying critical semiconductor components to ZTE, forcing the company to suspend major operations within weeks. Although the ban was lifted later that year, the episode was a powerful demonstration of how quickly access to semiconductors could become a binding constraint – effectively translating technological dependence into operational paralysis.

Together, these events heightened strategic mistrust on both sides, and reinforced Chinese officials' imperative to achieve technological self-reliance. As a result, the country intensified its efforts to localize semiconductor production and reduce its dependence on US technology, accelerating a broader trend toward supply chain decoupling. We will delve more into the specifics of China's present strategy later in the chapter.

¹⁴ [Semiconductors and the CHIPS Act: The Global Context," Congress.gov, Library of Congress.](#)

¹⁵ The Snowden revelations refer to the disclosure of thousands of classified documents from the U.S. National Security Agency (NSA) by Edward Snowden, a former NSA contractor. Beginning in June 2013, newspapers like The Guardian and The Washington Post published articles based on these documents, revealing the vast scale of global surveillance programs conducted by the NSA and its international partners, most notably the UK's Government Communications Headquarters (GCHQ).



2. The semiconductor supply chain: Complexity and chokepoints

Overcoming chokepoints in the semiconductor industry is more difficult than in the critical minerals sector. This is because the semiconductor ecosystem, while vast, is characterized by extreme specialization and a highly fragmented global value chain, particularly at the advanced end of the market.

Producing a single semiconductor is an extraordinarily complex process involving billions of transistors. This complexity is often explained by Moore's Law, first articulated by Intel co-founder Gordon Moore in 1965. Moore's Law states that the number of transistors on a chip doubles approximately every two years. At the same time, a complementary dynamic, often referred to as Moore's Second Law, dictates that the cost of manufacturing these chips rises exponentially as their components shrink.¹⁶ As transistors become smaller, fabrication becomes significantly more complex and capital-intensive. The combined effect of these two laws has driven a relentless cycle of innovation that requires ever-increasing R&D and capital expenditure.

This is reflected in the economics of production. A leading-edge fabrication plant (fab) can cost \$15–20bn to build, while industry leaders such as TSMC and Samsung spend \$30–40bn annually on capital expenditure alone. As one study notes for the semiconductor race, “the technological clock speeds up while the financial ante keeps rising.”¹⁷

2.1 Breaking down the global supply chain

The semiconductor supply chain can be divided into three main parts: (i) design, (ii) fabrication, and (iii) assembly, testing, and packaging (ATP). Each represents a distinct layer of value creation and a potential geopolitical chokepoint.

At the upstream end, the design stage involves creating the architecture and logic of a chip using highly specialized software and intellectual property (IP). This segment is dominated by US firms, particularly in electronic design automation (EDA) software, chip architecture, and core IP. As a result, the US maintains a critical advantage at the earliest stage of the value chain.

Midstream, the fabrication stage refers to the manufacturing of chips in foundries. This is the most capital-intensive and technologically complex part of the supply chain, with production highly concentrated geographically, particularly in Taiwan and South Korea. Taiwan dominates leading-edge fabrication—largely through TSMC—with a dense concentration of foundries producing chips for GPUs, CPUs, mobile devices, and connectivity systems. This dominance reflects decades of industrial execution, as well as powerful economies of scale inherent in advanced manufacturing, where costs and complexity favour a small number of highly specialized players.

However, even this dominance is constrained by a chokepoint in semiconductor manufacturing equipment (SME), particularly lithography. The Netherlands, through ASML, maintains a near-monopoly on extreme ultraviolet (EUV) lithography machines, which are required to produce leading-edge chips at 5nm and below. As a result, leading foundries such as TSMC depend on ASML's EUV systems to manufacture their most advanced semiconductors. Without these machines – which have direct implications for smartphones, data centers, and AI systems – production at the technological frontier would not be possible. This was demonstrated in 2021, when high demand exceeded ASML's production, leading to significant chip shortages and extended delivery times for advanced chips.

¹⁶ Brian Albrecht, Geoffrey Manne, David Teece, and Mario Zúñiga, “From Moore's Law to Market Rivalry: The Economic Forces That Shape the Semiconductor Manufacturing Industry,” International Center for Law & Economics, November 2025, <https://laweconcenter.org/wp-content/uploads/2025/11/ICLE-Moores-Law-FINAL-2025-11-12-1.pdf>.

¹⁷ Albrecht, Manne, Teece, and Zúñiga. “From Moore's Law to Market Rivalry”.



The durability of this chokepoint reflects immense technological and economic barriers to entry. Each EUV system costs over \$200mn and consists of highly complex, precision-engineered components, with no commercially credible alternative supplier. This dynamic reinforces the logic of Moore's Law: as performance improves, the cost and complexity of production rise sharply, making replication increasingly difficult. Even leading firms struggle to remain competitive—Intel's delays at the 10nm node in the late 2010s illustrate the difficulty of scaling advanced manufacturing processes and transitioning to next-generation lithography. Yet, at the same time, ASML's annual production cap of roughly 40-50 EUV systems effectively dictates the maximum possible scale of global AI compute, as approximately 3.5 EUV tools are required to produce one gigawatt of AI data center capacity.

At the downstream end, the assembly, testing, and packaging stage involves preparing finished chips for integration into final products. While less technologically complex, this stage is essential to the functioning of the broader ecosystem. Geographically, this stage is heavily concentrated in Asia, particularly in China and Southeast Asia. China alone accounts for roughly 30% of global assembly, testing, and packaging capacity, compared to just 4% in the US, according to the US-China Economic and Security Review Commission (USCC).¹⁸

Because these stages are rarely vertically integrated, production is typically split across firms. A fabless company designs the chip, a foundry manufactures it, and third-party providers handle the assembly, testing, and packaging. This has created a system in which different countries control different chokepoints: the US leads in design and software, Taiwan and South Korea in fabrication, and China in downstream manufacturing. Each stage is essential and difficult to replicate, making the system highly interdependent yet strategically fragile.

Figure 2: The global chip supply chain

Design	Stage 1: Intellectual property Design and license pre-designed components (IP cores) like processors, memory controllers, and interface protocols	Dominant players Arm Holdings, Synopsys, Cadence Design Systems, Imagination Technologies
	Stage 2: Electronic design automation EDA companies provide software tools, techniques, and methodologies for designing and producing electronic systems before physical manufacturing	Dominant players Cadence, Synopsys, Siemens, ANSYS, Keysight Technologies
Manufacturing	Stage 3: Fabless Fabless companies focus solely on the design and development of semiconductor chips, outsourcing the actual manufacturing to foundries	Dominant players NVIDIA, Qualcomm, Broadcom, AMD, Apple
	Stage 4: Foundries Specialised factories that manufacture chips for design firms. They turn the designs into physical chips and are a crucial part of the technology supply chain	Dominant players TSMC, UMC, Samsung, GlobalFoundries, SMIC
	Stage 5: Semi manufacturing equipment Produce the equipment crucial for the fabrication and testing of semiconductor devices	Dominant players Applied Materials, ASML, Lam Research, Tokyo Electron, KLA
Assembly, testing & packaging	Stage 6: Integrated device manufacturer Handle both the design and production of chips in-house, owning and operating their manufacturing facilities (fabs)	Dominant players Intel, Samsung Electronics, SK Hynix, STMicroelectronics, Micron
	Stage 7: Outsourced semiconductor assembly & test Specialise in providing third-party packaging and testing services for semiconductor manufacturers	Dominant players ASE Tech, Amkor Tech, JCET Group, Powertech, Tongfu Microelectronics

China US Taiwan Europe South Korea Japan

Source: Deutsche Bank Research.

¹⁸ "Chapter 9: Chained to China: Beijing's Weaponization of Supply Chains," in 2025 Annual Report to Congress, U.S.-China Economic and Security Review Commission, n.d.



2.2 A bifurcated industry: Advanced vs. legacy chips

Adding another layer to an already complex supply chain, semiconductors are also differentiated by their manufacturing process nodes, measured in nanometers (nm). These nodes reflect a chip's performance, efficiency, and technological sophistication. As the industry evolves, these nodes have effectively split the market into two parallel segments:¹⁹

- Advanced chips ($\leq 7\text{nm}$) → high-value, technologically complex, geopolitically sensitive
- Legacy chips ($\geq 22\text{nm}$) → high-volume, industrially essential, more geographically diffuse

In this age of infrastructure realism, control over advanced nodes provides technological leadership while dominance in legacy nodes secures a critical position in global manufacturing supply chains. At present, the US leads in advanced chip design and innovation, while China has built scale and dominance in legacy manufacturing. We delve into how the two countries have attained advantages below.

Advanced chips ($\leq 7\text{nm}$): The AI frontier

Advanced and leading-edge nodes—roughly 7nm and below—represent the technological frontier of semiconductor manufacturing. These chips power high-performance computing applications, including data centres, premium consumer devices, and artificial intelligence workloads, forming the backbone of modern digital infrastructure and cloud-based services.

The advent of AI has elevated the importance of advanced chips, transforming them from a market input into a strategic asset and a geopolitical chokepoint in recent years. Despite representing a small share of total semiconductor volumes, AI chips now drive a disproportionate share of industry growth, accounting for approximately 55% of the total chip revenue market by 2030 according to TSMC.²⁰

Thanks to its superiority in advanced compute, the US maintains a clear lead in AI. However, DeepSeek's launch in January 2025 challenged this perception and triggered a sharp ~17% decline in Nvidia's share price.²¹ This is because production of advanced and leading-edge chips is extremely capital-intensive and technologically complex. Thus, China's reveal of a high-performing and lower-cost model reportedly trained on less advanced hardware was enough to spook investors that demand for their increasingly more costly hardware would taper off.

Despite the DeepSeek scare, China's AI capabilities remain constrained by access to advanced semiconductors. DeepSeek itself relied on Nvidia H800 chips—which are subject to US export controls but still based on US architecture—to train its models, reportedly using 2,788,000 GPU-hours (equivalent to around \$5.6mn, assuming \$2 per GPU-hour).²² Investigations by CSIS also suggest that restricted H100 chips may also have been accessed indirectly by DeepSeek to train its model.²³ According to Epoch AI, Chinese large language models (LLMs) lag their US counterparts by approximately seven months.²⁴

¹⁹ These categories (referring to chip nodes/process nodes) are a subjective definition and reflect our views, not a standard defined by the industry.

²⁰ Company, Manufacturing. 1970. "2026 Technology Symposium Highlights." Tsmc.com. Taiwan Semiconductor Manufacturing Company. 1970. https://www.tsmc.com/english/symposium_highlights/2026.

²¹ According to Nvidia CEO Jensen Huang, AI competition is a "five-layer cake": power, chips, infrastructure, models, applications. The US leads in chip design (Nvidia), infrastructure (hyperscalers) and frontier models (Anthropic, OpenAI), while China leads in efficiency and open-source.

²² GPU rental prices were based off CSIS calculations from 2025; rental prices for GPUs per hour have since grown exponentially as of May 2026, ranging from \$3.50-\$10 GPU/hour. Gregory C. Allen, "DeepSeek, Huawei, Export Controls, and the Future of the U.S.-China AI Race," Center for Strategic and International Studies, 2025, <https://www.csis.org/analysis/deepseek-huawei-export-controls-and-future-us-china-ai-race>.

²³ Allen, "DeepSeek, Huawei, Export Controls."

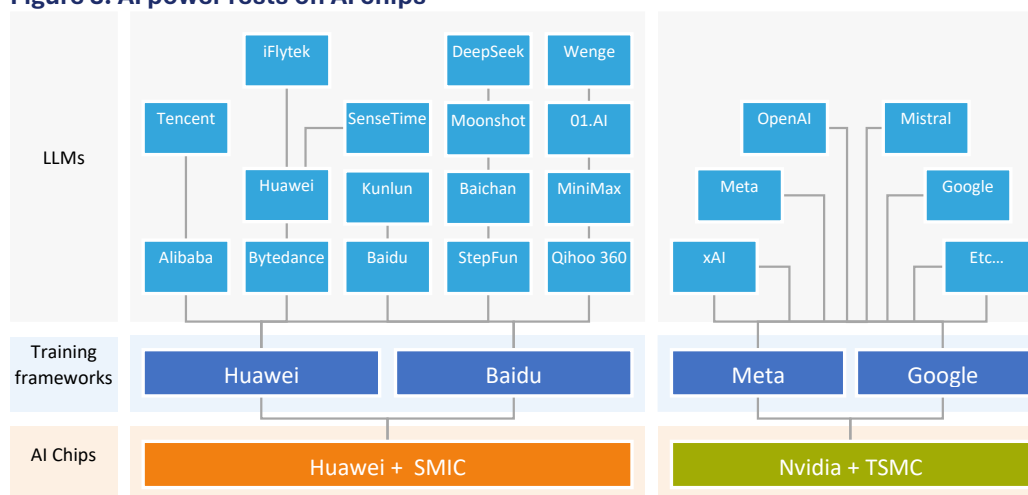
²⁴ Luke Emberson, "Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023," Epoch AI, 2023, <https://epoch.ai/data-insights/us-vs-china-eci/>.

The broader issue is that China remains compute-constrained to produce any of its own leading-edge chips. A key bottleneck lies in high-bandwidth memory (HBM), which provides the storage within AI chips that can help train and serve large AI models. HBM requires 4x the wafer space of regular memory chips, and according to Epoch AI, AI chips already consumed more than 90% of the HBM industry's output in terms of revenue in 2025.²⁵ And feeding the AI boom will require even greater dependence on ASML, which China does not have access to, as HBM is built from DRAM wafers produced with its EUV tools – which, as mentioned earlier, have a finite annual production capacity. While China's ChangXin Memory Technologies (CXMT) is making progress, global leaders like SK Hynix, Samsung, and Micron remain more than two generations ahead.

In addition, Chinese chip ecosystems are plagued by weaker software integration and stability issues. Domestic alternatives from firms like Huawei and Cambricon are less competitive, with some estimates suggesting performance levels of around 80% of Nvidia's A100.²⁶ As a result, China's AI development continues to depend on Western hardware and architectures. In contrast, the US is buoyed by massive capital investment from its hyperscalers, absolute dominance in frontier software models, and a top-tier global AI research talent pool.

Finally, export controls implemented by the US and its allies on advanced semiconductor equipment—particularly EUV lithography tools—also continue to limit China's ability to scale leading-edge production. Under Moore's Law, the compute requirements for frontier AI models continue to rise rapidly, further widening this gap.

Figure 3: AI power rests on AI chips



Source: Mercator Institute for China Studies, Deutsche Bank Research.²⁷

Legacy / mature chips ($\geq 22\text{nm}$): Scale and industrial resilience

In contrast, legacy and mature nodes—typically $\geq 22\text{nm}$ —form the backbone of the physical economy. These chips are widely used in automotive systems, power management, sensors, microcontrollers, and industrial machinery, as well as consumer

²⁵ "Total AI Chip Memory Bandwidth Has Grown 4.1x per Year, Now Reaching 70 Million Terabytes per Second," Epoch AI, March 24, 2026, <https://epoch.ai/data-insights/hbm-shipped/>.

²⁶ Chris McGuire, "China's AI Chip Deficit: Why Huawei Can't Catch Nvidia and U.S. Export Controls Should Remain," Council on Foreign Relations, December 15, 2025, <https://www.cfr.org/articles/chinas-ai-chip-deficit-why-huawei-cant-catch-nvidia-and-us-export-controls-should-remain>.

²⁷ Wendy Chang, Rebecca Arcesati, and Antonia Hmaid, "MERICS Report: China's Drive Toward Self-Reliance in Artificial Intelligence from Chips to Large Language Models," July 2025, https://merics.org/sites/default/files/2025-07/MERICS%20Report-AI_Stack_final.pdf.



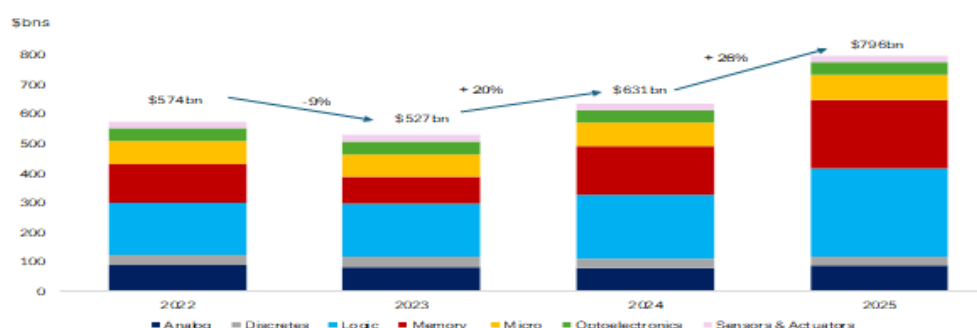
electronics and defence applications.²⁸ They are characterized by lower production costs, high reliability, and long product life cycles.

Within this category, mature nodes represent a critical middle layer, balancing performance and cost. These chips are prevalent across automotive and industrial applications, as well as mainstream consumer electronics. Production is widely distributed across global players such as TSMC, UMC, SMIC, and GlobalFoundries.

China has emerged as a dominant player in this segment. Since 2014, it has been able to significantly expand its manufacturing capacity through sustained state subsidies, particularly via firms such as SMIC, Hua Hong, and Nexchip, and by focusing on mature-node manufacturing production, where it possesses the necessary expertise and equipment.²⁹ According to the Semiconductor Industry Association (SIA), China is projected to account for roughly 35–40% of global mature-node capacity by 2027.³⁰

This expansion is already reshaping global markets. Increased Chinese supply has contributed to pricing pressure on international foundries, and some global firms are increasingly sourcing from Chinese manufacturers due to cost advantages—for example, in PC supply chains involving companies such as HP, Dell, and ASUS.³¹

Figure 4: Semiconductor shipments by segment (\$ bn)



Source: Deutsche Bank Research, World Semiconductor Trade statistics.

3. The geopolitical chessboard: Taiwan's central role

No discussion of the US-China chip competition is complete without spotlighting Taiwan's pivotal role in the race. Taiwan produces over 90% of the world's most advanced semiconductors and roughly around 55% of the world's semiconductors, making it a critical chokepoint in the global technology supply chain. US Commerce Secretary Howard Lutnick has vowed to reduce dependence on Taiwan's semiconductors as quickly and as much as possible, calling for up to 40% of Taiwan's-linked semiconductor supply chains to eventually be reshored to the US.³²

3.1 Global dependence on a single island

Taiwan is a high-tech economy deeply embedded in global supply chains. Exports account for roughly 70% of its GDP, with semiconductors acting as the primary engine of growth.³³

²⁸ Process nodes refer to the feature size of the transistors on a semiconductor. Advanced logic chips are made below the 22-nm threshold, where node shrinkage generally yields faster and more efficient chips but requires more complex manufacturing processes. Within advanced, cutting-edge chips use 3-nm and 5-nm process modes.

²⁹ "Public Report on the Use of Mature-Node Semiconductors," Bureau of Industry and Security, December 2024, <https://www.bis.gov/media/documents/public-report-use-mature-node-semiconductors-december-2024.pdf>.

³⁰ "Winning the Chip Race," Semiconductor Industry Association, 2025.

³¹ Lauly Li and Cheng Ting-Fang, "HP, Dell, Acer and Asus Mull Using Chinese Memory Chips amid Supply Crunch," *Nikkei Asia*, February 5, 2026, https://asia.nikkei.com/spotlight/supply-chain/hp-dell-acer-and-asus-mull-using-chinese-memory-chips-amid-supply-crunch?utm_source=substack&utm_medium=email.

³² Dylan Butts, "Impossible": Taiwan Pushes Back against Washington's 40% Chip Supply Relocation Goal," *CNBC*, February 10, 2026, <https://www.cnbc.com/2026/02/10/taiwan-chips-us-supply-chain-lutnick-trade-deal.html>.

³³ Calculated using Haver data.



At the centre of this ecosystem is Taiwan Semiconductor Manufacturing Company (TSMC), which alone produces 90% of the world's most advanced chips.

The US remains heavily reliant on Taiwan for chips, despite its ongoing efforts to build domestic capacity through the 2022 CHIPS and Science Act and by applying political pressure on TSMC to expand its US footprint. In 2025, 70% of TSMC's revenue came from the US, compared to 10-15% from China.³⁴

Trade flows underscore this dependence. In 2025, Taiwan was the US's fifth-largest trading partner and fifth-largest source of imports, resulting in a US trade deficit with Taiwan of \$147bn.³⁵ This reliance is particularly acute in the AI sector. Nvidia – now the world's most valuable company with a market cap that exceeds the entire stock market value of several G7 nations – depends almost entirely on TSMC to produce its advanced GPU and AI chips. These chips account for most of Nvidia's data centre revenue, its fastest-growing segment.

At the same time, Taiwan remains intrinsically linked to China. The country is Taiwan's largest investment and trading partner, even though exports to China have halved in the last few years, from 42% in 2021 to 23% in 2025.³⁶ While Taiwan has sought to decouple from China, the latter remains the largest single destination for Taiwan's chip exports by value. This is because China serves as the world's final-assembly hub, integrating these components into electronics that are then re-exported globally, tying China's manufacturing ecosystem inextricably to Taiwan's foundries.

3.2 How did Taiwan (and TSMC) become so powerful?

Today, TSMC is Taiwan's prized industrial jewel, owing much of its success to an early-mover advantage in semiconductor manufacturing in the late 1980s. Its rise was supported by close ties to Silicon Valley – many of its first engineers were trained in the US – as well as significant Taiwan government funding, which provided almost half of its initial capital.³⁷ This occurred as US power reached its post-Cold War peak, championing a globalized order built on the belief that expanding trade and integrated supply chains promoted peace and economic prosperity. The resulting mutual interdependence between TSMC and Silicon Valley was the best example of this. As a result, officials from both the public and private sectors saw no issue in increasingly outsourcing their chip designs to fab companies abroad such as TSMC.

Crucially, TSMC pioneered the pure-play foundry model. By manufacturing chips designed by external customers, it enabled the rise of the US's "fabless" semiconductor giants, such as Qualcomm, Nvidia and Apple. This allowed these companies to focus on high-value design while outsourcing capital-intensive manufacturing, a division of labor that fundamentally reshaped the global semiconductor industry.

TSMC also distinguished itself through a willingness to take calculated technological risks. Its pivotal partnership with Apple in the early 2000s – after other firms had declined to manufacture chips for the first iPhone – helped cement its position as a leading-edge producer. TSMC was also an early and committed adopter of extreme ultraviolet (EUV) lithography in the 2010s, despite significant uncertainty surrounding the technology at the time. More recently, it correctly anticipated the need to expand advanced packaging capacity to support Nvidia's rapid rise in the AI chip market.

The ability to "anticipate technological inflection points" has been central to TSMC's success.³⁸ Its premier customer base has enabled it to continuously invest in increasingly advanced nodes, despite the exponentially rising costs associated with each new

³⁴ "TSMC Annual Report 2024," Taiwan Semiconductor Manufacturing Company, 2025, <https://investor.tsmc.com/sites/ir/annualreport/2024/2024%20Annual%20Report-E.pdf>.

³⁵ Karen M. Sutter, "U.S.-Taiwan Trade and Economic Relations," CRS In Focus IF10256 (Washington, DC: Congressional Research Service, 2026), <https://www.congress.gov/crs-product/IF10256?h=semiconductors&s=1&r=1>.

³⁶ Sutter, "U.S.-Taiwan Trade."

³⁷ Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (New York: Simon and Schuster, 2022).

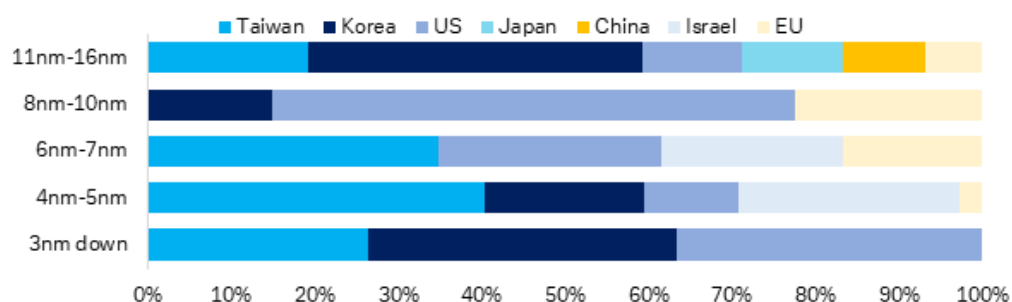
³⁸ Albrecht, Manne, Teece, and Zúñiga, "From Moore's Law to Market Rivalry."



generation of semiconductor manufacturing. As a result, TSMC recently joined the elite group of nine companies globally that enjoy a market cap above one trillion dollars – and this despite the tariff-related uncertainty stemming from the US's 15% tariff levied recently against Taiwan.

Finally, Taiwan's semiconductor strength extends beyond TSMC. Other major local players include United Microelectronics Corporation (UMC), Nanya Technology Corporation, and Winbond Electronics, which rank among the world's leading manufacturers of memory and specialty chips.

Figure 5: High-end node fabrication capacity by region/island (nanometres)



Source: Deutsche Bank Research, Bloomberg Finance LP (2021 data). Estimates are based on the machinery installed, and refer to theoretical capacity, not what has been commercially produced. The data also refers to all semiconductor types.

3.3 Mutually assured dependence

Despite its market dominance, TSMC does not operate independently of the broader semiconductor ecosystem. Instead, it is deeply embedded in a system of mutual reliance. The TSMC-Apple relationship illustrates this dynamic best: TSMC depends on large, predictable orders from Apple to justify its massive capital expenditures, while Apple depends on TSMC's cutting edge manufacturing capabilities to maintain its technological edge. TSMC has replicated this model with other major clients, including Nvidia, AMD, and Broadcom. This symbiotic relationship underscores the Taiwan and US relationship of mutually assured dependence, where the fortunes of both are so closely intertwined that the relationship itself has increasingly become a source of strategic vulnerability.³⁹

Historically, TSMC has also acted as a bottleneck that has kept the China/US competition asymmetric. Until 2020, Huawei was TSMC's second-largest customer, relying on it for advanced chip fabrication. However, this shifted following the US expansion of the Foreign Direct Product Rule (FDPR) in May 2020 – an extraterritorial measure that restricted any company using US-origin technology from supplying Huawei. Given TSMC's reliance on US semiconductor equipment and design software, the company determined it was required to comply with the new regulation, leading it to halt shipments to Huawei by September 2020. As a result, Huawei's sales collapsed, declining 30% in H1 2021, forcing it to stockpile aggressively and sell off divisions to conserve its chips for strategic priorities.⁴⁰

The rapid growth of AI has only reinforced this dependence, increasing demand for advanced chips and further embedding Taiwan within US supply chains. With TSMC holding a monopoly on the most advanced sub-3nm nodes, Taiwan effectively dictates the extent of the US's AI progress and, by extension, its national security strategies. Despite the more isolationist rhetoric of the second Trump administration, Taiwan's share of US high-tech imports has continued to rise, underscoring the structural nature of this reliance.

³⁹ Albrecht, Manne, Teece, and Zúñiga, "From Moore's Law to Market Rivalry."

⁴⁰ BBC News. 2021. "Huawei Revenue Sees Biggest Ever Fall," August 6, 2021, sec. Technology. <https://www.bbc.co.uk/news/technology-58104614>.



TSMC's early conservatism in scaling up for AI growth also created a new bottleneck, with forecasts suggesting advanced compute may not meet AI-driven demand until 2028 or 2029.⁴¹ Notably, the company chose to keep its capex relatively flat during 2023 and 2024, just as the AI boom was accelerating post-ChatGPT.⁴² Given the 2-3 year lead time required to build a new semiconductor fab, TSMC's conservatism has created a significant supply/demand gap that is not expected to narrow significantly over the next couple of years. This constraint has been noted in recent earnings calls by leading hyperscalers and has incentivized both the US government and the private sector to invest in domestic foundry competition, particularly through Intel and Samsung in an attempt to diversify.

Finally, TSMC remains embedded in China's supply chains. Although the company has attempted to move away from its reliance on mainland China, it still reports that 9-10% of its revenue comes from Chinese customers. Furthermore, exposure remains high, as overall non-US revenue is around 80%.⁴³

Ultimately, while the US still leads the global technology race, the TSMC chokepoint means that its leadership is defined more by resource management than absolute abundance. This dynamic critically impacts US strategy toward China.

3.4 Protecting the silicon shield

TSMC also acts as a bottleneck to the US's AI progress. This is because AI has a critical physical dependency – specifically, on the fabrication plants of TSMC in Taiwan.⁴⁴ This deep reliance on Taiwan's semiconductor supply suggests that the US might hesitate to relinquish its security commitment to Taiwan, even amid efforts to stabilize relations with China.

The economic consequence of any disruption to Taiwan's semiconductor supply could be severe. A blockade or conflict scenario could trigger a global shortage of advanced chips, effectively paralyzing large parts of the global economy. Bloomberg Economics estimates that a Chinese blockade of Taiwan could result in \$5trn in economic losses in the first year alone.⁴⁵ Similarly, the Semiconductor Industry Association warns that a complete disruption could lead to the "largest economic crisis since the Great Depression," with US GDP potentially declining by 11% and output decreasing by 16%.⁴⁶

Given these stakes, the ongoing tensions between China and Taiwan regarding Taiwan's sovereignty have heightened global concerns about supply chain security. In response, the US, Japan, and the EU have sought to diversify semiconductor manufacturing capacity. However, replicating Taiwan's ecosystem would take years, if not decades, and would come at a significantly higher cost.

Ultimately, as the sole provider of the most advanced nodes, TSMC's manufacturing capacity and its geographic safety in Taiwan plays a key role in the limits of US national security power and AI progress.

⁴¹ The extent of TSMC's hegemony in the chip ecosystem was spotlighted in August 2025, when news broke that several employees had stolen critical information on TSMC's 2-nanometre chip technology, marking the first-ever case involving national core sensitive technologies to be investigated under Taiwan's National Security Act, enacted in 2022 to protect its chip industry from foreign threats. See: Cheng Ting-Fang and Lauly Li, "Taiwan Indicts Trio for Alleged Theft of TSMC's 2-Nm Trade Secrets," *Nikkei Asia*, August 28, 2025, <https://asia.nikkei.com/business/technology/taiwan-indicts-trio-for-alleged-theft-of-tsmc-s-2-nm-trade-secrets>.

⁴² Ben Thompson, "TSMC Earnings, the TSMC Brake Revisited, Why AI Needs Foundry Competition," *Stratechery*, January 21, 2026, <https://stratechery.com/2026/tsmc-earnings-the-tsmc-brake-revisited-why-ai-needs-foundry-competition/>.

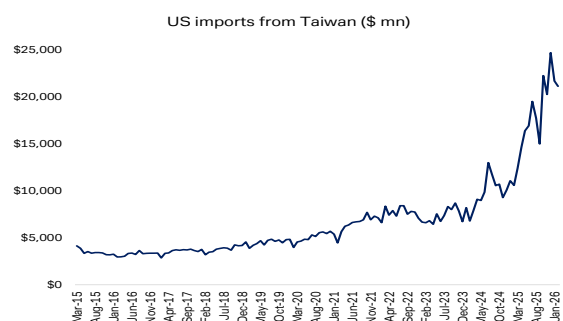
⁴³ TSMC 2025 annual report.

⁴⁴ Thompson, "TSMC Earnings."

⁴⁵ Chris Anstey, "Bloomberg New Economy: China 'Blockade Simulation' Raises \$5 Trillion Risk," Bloomberg.com, May 25, 2024, <https://www.bloomberg.com/news/newsletters/2024-05-25/bloomberg-new-economy-china-blockade-simulation-raises-5-trillion-risk>.

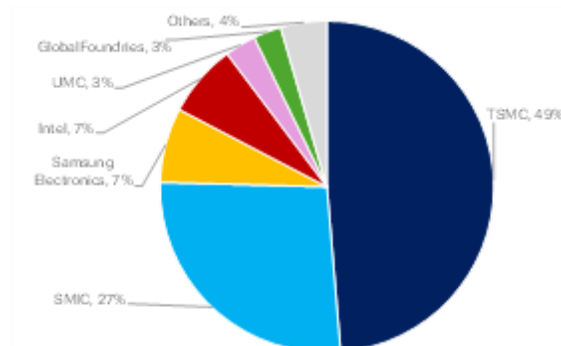
⁴⁶ Mickle, "The Looming Taiwan Chip Disaster."

Figure 6: The AI race has fuelled chip demand, pushing up Taiwan's share of US imports over the past few years



Source: Census Bureau, Haver Analytics.

Figure 7: TSMC continues to dominate in foundry capacity, keeping a cap on global AI progress



Source: Bloomberg Finance LP, Deutsche Bank Research.

4. US vs. China: A shifting balance

The strategic vulnerability posed by TSMC illustrates a broader structural weakness in the US semiconductor ecosystem. Once the undisputed pioneer in the chip race, today's US companies are dependent on global partners for manufacturing, assembly and packaging. The decline is stark: in 1990, the US accounted for nearly 40% of global chip production but by 2019, that share had plummeted to just ~11%.⁴⁷

4.1 The US response: Rebuilding domestic capacity

Since 2022, successive US administrations have sought to strengthen their semiconductor supply chains. The most significant effort thus far has been the ~\$278bn CHIPS and Science Act, a landmark piece of legislation designed to bolster domestic chip capabilities via direct subsidies and tax credits for manufacturing. As of December 2025, the Department of Commerce has awarded ~\$30.9bn across 40 projects to 19 companies.⁴⁸ These include massive, multi-billion-dollar packages to foreign companies like TSMC (\$6.6bn), Samsung (\$4.75bn) and Intel (\$7.86bn), all part of the US government's strategy to onshore advanced chip manufacturing.

Under CHIPS, advanced semiconductors are now being made in the US for the first time in over a decade at TSMC's new Arizona fab. The core theory behind this is that if the US could onshore anchor tenants like TSMC, Samsung in Austin, Intel in Arizona, and SK Hynix in Indiana, a domestic ecosystem of suppliers would naturally follow. While the CHIPS Act represented the first clear strategic vision and effort at rebuilding the US's domestic chip supply chains, its success in weakening China's long-term industrial power and strengthening its own supply-chain has been mixed.

First, the \$278bn funding, while substantial, could not outcompete Moore's Law, and proved insufficient for total de-risking. Within the overall funding for example, the \$39bn allocated to revive domestic semiconductor manufacturing pales in comparison to China's own state-led spending, which totals around \$142bn – roughly 3x the total funding the US has allocated towards direct industrial support.⁴⁹ Second, the Act's limited resources led to the neglect of several critical downstream segments of the US supply

⁴⁷ Blevins, Platzer, and Sutter, "Semiconductors."

⁴⁸ "Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain," U.S. Government Accountability Office, GAO-26-107882, December 11, 2025, <https://www.gao.gov/products/gao-26-107882>.

⁴⁹ [China has spent 3.6 times more than the US on chipmaking subsidies over the past decade — \\$142 billion and counting, easily outweighs CHIPS Act | Tom's Hardware](#)



chain system like conventional packaging and materials. Third, CHIPS has faced difficulty attracting suppliers. Outside of the giants like TSMC and Intel, smaller Asian suppliers have been hesitant to onshore operations until more fabs or sufficient volume exist to justify the shift from exporting from Taiwan.⁵⁰ Finally, the initiative suffered from a lack of formal coordination amongst officials to protect and promote the US chip supply chain, with few clear metrics for what “self-sufficiency” would look like.

The limited success of the US CHIPS Act’s implementation suggests the US can no longer scale industrial projects as rapidly as it did during the 20th century.⁵¹ Under the second Trump administration, the US has reportedly revisited some of the multi-billion-dollar contracts, although the CHIPS program office has been transferred to the newly created “US Investment Accelerator” within the Department of Commerce, tasked with fulfilling pre-existing awards. Interestingly, the administration has also expanded on what the CHIPS Act money is directed towards, with new investments in areas like critical minerals and materials now being considered.⁵²

How has the second Trump administration tried to solve US chip dependency on Taiwan? The slew of trade deals negotiated in 2025 all contain provisions requiring partner nations – including Japan, Malaysia, and Taiwan –to invest in core US industries such as semiconductors. The January 2026 US-Taiwan Agreement on Trade and Investment is particularly notable for its focus on semiconductors.⁵³ Among the provisions, Taiwanese chip producers that invest in the US will get a cut on tariff imports of up to 2.5x the planned capacity. US Secretary of Commerce Howard Lutnick has said that the administration’s objective is to bring 40% of Taiwan’s chip supply chain and production onshore.⁵⁴

This strategy has been backed by some dramatic interventions regarding tariffs. For example, in 2025 President Trump threatened TSMC with tariffs of up to 100% if it did not expand its US presence.⁵⁵ In return, the company pledged a total \$165bn investment in the US as it sought to quell accusations that Taiwan was undermining the US semiconductor business.⁵⁶

Leading US companies are also attempting to address Taiwan’s bottleneck in AI chips. Nvidia, for instance, is transitioning from a monolithic GPU to a “multi-architectural” approach, designing multiple specialized chips to handle different AI workloads.⁵⁷ In doing so, Nvidia aims to ensure that no component of the AI compute stack becomes a bottleneck that leaves expensive GPUs idle.

4.2 Cutting off the chokepoints

In addition to the domestic measures taken to strengthen its chip resilience, since 2010 the US has also implemented a series of export controls designed to manage the flow of semiconductor technology to China. Its measures on China were initially focused on specific companies like Huawei and SMIC. As trade tensions and allegations of espionage and intellectual property theft grew, the Department of Commerce added Huawei to its

⁵⁰ Jordan Schneider and Phoebe Chow, “The Future of Economic Security with Dan Kim and Chris Miller,” *ChinaTalk* (podcast), January 21, 2026, <https://www.chinatalk.media/p/the-future-of-economic-security-with->

⁵¹ Trump has also turned to domestic efforts like The Pax Silicon Fund, a US-led multilateral coalition that includes an investment fund targeting \$4trn in funding for energy projects, critical minerals and semiconductor supply chains. The US has announced it will contribute \$250mn toward the fund.

⁵² “Tracking CHIPS and Science Act Awards,” *Manufacturing Dive*, December 5, 2024, updated March 6 2026, <https://www.manufacturingdive.com/news/chips-and-science-act-tracker-semiconductor-manufacturing/734039/>.

⁵³ “Fact Sheet: Restoring American Semiconductor Manufacturing Leadership through an Agreement on Trade & Investment with Taiwan,” U.S. Department of Commerce, Office of Public Affairs, January 15, 2026. <https://www.commerce.gov/news/fact-sheets/2026/01/fact-sheet-restoring-american-semiconductor-manufacturing-leadership>.

⁵⁴ “Howard Lutnick Breaks down Recent \$250B Trade Deal with Taiwan,” CNBC Television, YouTube, January 15, 2026, <https://www.youtube.com/watch?v=cSoPNbdLLTg>.

⁵⁵ “Trump Says He Told TSMC It Would Pay 100% Tax If It Doesn’t Build in US,” *Reuters*, April 9, 2025, <https://www.reuters.com/world/us/trump-says-he-told-tsmc-it-would-pay-100-tax-if-it-doesnt-build-us-2025-04-09/>.

⁵⁶ “TSMC Intends to Expand Its Investment in the United States to US\$165 Billion to Power the Future of AI,” Taiwan Semiconductor Manufacturing Company Limited, March 4, 2025, <https://prtsmc.com/english/news/3210>.

⁵⁷ Pratik Kumar Patel, “Optimize Data Center Efficiency for AI and HPC Workloads with Power Profiles,” NVIDIA Technical Blog, December 4, 2025, <https://developer.nvidia.com/blog/optimize-data-center-efficiency-for-ai-and-hpc-workloads-with-power-profiles/>.



entity list in 2019, which prevented the company from obtaining advanced chips developed or produced in the US. In 2020, Commerce added SMIC to the entity list.⁵⁸ The regulations concerning Huawei were further expanded that year when the US modified its Foreign Direct Product Rule (FDPR) to prevent foreign fabs like TSMC from supplying certain chips to Huawei and other listed entities without US authorization. The focus on specific firms like Huawei was intended to limit China's influence at the end-user stage of the supply chain, where the country held a leading role in global markets.

Washington has been able to exert this pressure because, as we noted earlier, US companies dominate upstream segments of the chip supply chain, including fabless design, advanced electronic design automation (EDA) software, and critical IP. This dominance underpins the “extraterritorial” reach of US export controls, where any chip made with US tools or IP can be legally restricted from being sold to Chinese companies like Huawei.⁵⁹

The FDPR's 'extraterritorial' reach created a clear choice for global foundries in 2020: they could either continue using US technology or continue supplying designated entities like Huawei, but not both. As TSMC's operations relied on US software and machine tools/equipment to power its foundries (illustrating the “mutually assured dependence” relationship introduced earlier), the company suspended production of Huawei's Kirin chips, and related services from companies like Google were withdrawn. As a result, Huawei's smartphone business—once shipping >200m phones/year—collapsed in the high-end segment.

Washington has also implemented measures directed at the manufacturing segment of China's semis supply chain. In 2019, the US put restrictions in place to prevent ASML from selling advanced EUV equipment to China, further tightening these in 2023 to include the selling of advanced EUV and related components to China. The US also added SMIC to the entity list and restricted its access to tools that could create 10nm and below chips in 2020.

As the global chip industry recuperated from the pandemic, US export control policy shifted more broadly to address China's development of cutting-edge chips. In a landmark ruling in October 2022, the US Department of Commerce imposed controls on exports to China of advanced logic AI chips below defined performance thresholds of 16nm/14nm.⁶⁰

Commerce's rules also restricted the country's chief chip designer Nvidia from selling its flagship A100 and H100 GPUs to China. In response, Nvidia developed modified A800 and H800 chips for the Chinese market, but these too were later controlled by the Department of Commerce's Bureau of Industry and Security (BIS) in November 2023.⁶¹ Nvidia's subsequent H20 chip, designed to comply with the updated control thresholds, was a source of contention between the company and the US government until August 2025, when the government approved its sale to China in return for 15% of sales revenue from Nvidia.

As the AI race gathered steam, Japan and the Netherlands joined the US in restricting the export of advanced equipment to China in January 2023.⁶² Other successive actions during this period include the addition of 140 more Chinese companies to the Entity List in

⁵⁸ “Commerce Department Further Restricts Huawei Access to U.S. Technology and Adds Another 38 Affiliates to the Entity List,” U.S. Department of Commerce, August 17, 2020, <https://2017-2021.commerce.gov/news/press-releases/2020/08/commerce-department-further-restricts-huawei-access-us-technology-and.html>.

⁵⁹ Marion Labouré and Cassidy-Ainsworth Grace, “Thematic Research: CHARTBOOK - Tangled Wires: The Geopolitical Landscape of Semiconductors & Rare Earth Metals,” Deutsche Bank Research, June 2023, https://research.db.com/research/Article?rid=e5a16334_9484_490f_9217_7ab2649de709_604&id=RP0001&wt_cc1=IND-3146993-0000.

⁶⁰ Demetri Sevastopulo and Kathrin Hille, “US Hits China with Sweeping Tech Export Controls,” *Financial Times*, October 7, 2022, <https://www.ft.com/content/6825bee4-52a7-4c86-b1aa-31c100708c3e?syn-25a6b1a6=1>.

⁶¹ Karen M. Sutter, 2025, “U.S. Export Controls and China: Advanced Semiconductors,” CRS Report R48642 (Washington, DC: Congressional Research Service, 2025), <https://www.congress.gov/crs-product/R48642>.

⁶² Demetri Sevastopulo and Sam Fleming, “Netherlands and Japan Join US in Restricting Chip Exports to China,” *Financial Times*, January 28, 2023, <https://www.ft.com/content/baa27f42-0557-4377-839b-a4f4524cfa20?syn-25a6b1a6=1>.



December 2024, the expansion of the FDPR to include High-Bandwidth Memory (HBM), and further controls on advanced SME equipment for leading-edge logic and memory nodes.

Figure 8: Nvidia's modified chips for China have evolved along with the US government's policies

Date	US Bureau of Industry and Security (BIS) export controls actions	Nvidia's modified chips and related responses
Oct-2022	A100 and H100 chip export controls	Announced A800 and H800 modified chips for China
Nov-2023	A800 and H800 export controls	Announced H20, L20 and L2 modified chips for China
May-2025	Controlled the H20	Announced efforts for B30 and B40 for China
Jul-2025	Grants export licenses for H20 sales in China	China's Cybersecurity administration expressed concerns regarding H20
Dec-2025	Approved export licenses on a case-by-case basis for Nvidia's H200 and AMD's M1235X AI logic devices	Chinese customs officials initially said H200 chips were "not permitted" to enter China, later loosened to accept limited imports. Restrictions by both governments (the Trump admin has also placed a 25% tariff on H200 chip imports, which is deterring Nvidia

Source: Deutsche Bank Research, Congressional US reports, CSIS.

4.3 The Trump administration and the turning point

The current US policy towards China on semiconductors has become more uncertain and, at times, appears contradictory. The Trump administration has not sought to unilaterally deny China access to technology in the same manner as the 2022–24 framework under the Biden administration. Instead, its approach has involved a partial relaxation of chip controls, seemingly in favour of monetizing and re-politicizing the chokepoints.

The biggest demonstration of this shift occurred in December 2025, when the Trump administration granted permission for Nvidia to sell its second-best AI chip, the H200, to China in exchange for a 25% share in revenue. This was a notable step away from the 15% cut in the less-advanced H20 chip deal in August. Still, the precise mechanism – whether a security check before being shipped to China, followed by a tariff or another process to collect the 25% – remains unclear. Following this decision, in early 2026, Trump imposed a 25% tariff on advanced chips with total processing performance of less than 21,000 and total DRAM bandwidth of less than 6,500 GBs, and their spin-off products used in AI applications. Nevertheless, these moves mark a departure from the hardline stance of previous administrations and are a dial-down from the 100% rate on chip imports that Trump first indicated the US would impose in August 2025.⁶³

The administration's more accommodative stance was further demonstrated in February, when the White House overruled a Department of Defense (DoD) recommendation to blacklist major Chinese firms, subsequently removing companies like Alibaba and Baidu from the sanctions list.⁶⁴

The Trump administration's seemingly contradictory policy can be understood as the result of several competing pressures. On one hand, some experts believe the policy change was driven by concerns over the declining effectiveness of export controls – evidenced by the rise of competitors like Huawei.⁶⁵ Others are of the opinion that President Trump is seeking diplomatic stability with Beijing in 2026, focusing on cooperation on issues like fentanyl and soybean purchases. A more prevalent view is that the administration is divided internally between a very hawkish camp that advocates for strict controls on China, versus the diffusionist camp, led by White House AI Czar David Sacks and the lobbying influence of Nvidia, who argue that, in order to promote US AI leadership, US tech standards should be diffused as quickly as possible before a Chinese

⁶³ Anthony Rapa and Rachel D Evans, "Trade Rules: Trump Administration Revises Export Restrictions on Advanced Semiconductors While Levying Additional Tariffs," *Reuters*, March 13, 2026, <https://www.reuters.com/legal/legalindustry/trade-rules-trump-administration-revises-export-restrictions-advanced-pracin-2026-03-13/>.

⁶⁴ Edwin Chan, "Alibaba Leads Tech Slide after Pentagon Briefly Shows Blacklist," *Bloomberg.com*, February 16, 2026, <https://www.bloomberg.com/news/articles/2026-02-16/alibaba-leads-tech-slide-after-pentagon-briefly-shows-blacklist>.

⁶⁵ Chris McGuire, "China's AI Chip Deficit."



competitor like Huawei can catch up and lead the world to build AI capabilities on the Huawei stack.⁶⁶

Once again, Taiwan stands as the central chokepoint. As the sole provider of the most advanced nodes, TSMC's manufacturing capacity and its geographic safety in Taiwan ultimately dictate the limits of US national security power and AI progress. The US cannot completely isolate China from the advanced chip supply when it does not control the primary manufacturing gateway. To maintain its lead, the US must ensure its tech stack diffuses globally to gain a competitive advantage before it can effectively exercise control, particularly as China excels in model training (like DeepSeek) and dominates the open-source landscape.

A final consideration in this strategic calculus is China's control over critical minerals, which provides the central government with significant counter-leverage. The retaliatory dynamic between the US and China on critical minerals and semiconductors has led some experts to conclude that China has a greater willingness to absorb economic pain to achieve strategic objectives than the US. By implementing its own export restrictions on rare earths – a move that was not without cost to its own economy – China demonstrated a capacity to retaliate, potentially influencing the US to de-escalate its own measures.⁶⁷

In this context, perhaps the only short-term solution is for the US to continue loosening its export controls on advanced chips to China, which the Trump administration is already doing. This decision has split stakeholders, with some corporate officials arguing that doing so is necessary to keep China dependent on the US AI stack, while other public and government officials warn that this approach provides Beijing with the tools to eventually outcompete the US in the global race for AI hegemony. For example, quantitative analysis from the Council on Foreign Relations (CFR) projects that if the US exports 3mn H200 chips to China in 2026, it would give China more AI computing power than it could produce domestically until 2028 or 2029 earliest.⁶⁸

As a result, this policy has faced backlash from congressional hawks who oppose the new AI chip licenses to China, with the Financial Times reporting that several bipartisan officials called on the Department of Commerce to suspend Nvidia's export licenses to China.⁶⁹ Congressional lawmakers are also looking to close loopholes in Trump's more relaxed posture towards Chinese tech via the proposed Remote Access Security Act, which will extend chip export controls to cover Chinese access to restricted compute via overseas cloud services.

A major proposal, the MATCH (Multilateral Alignment of Technology Controls on Hardware) Act, would allow the US to apply FDPR rule against allies if they do not match US chip export controls. This Act specifically targets technologies like ASML's EUV equipment and, if passed, could significantly increase the risk of Chinese retaliation. However, many investors we have spoken to see this as more of a strategic move than a real threat, fearing China would retaliate with rare earth export controls, as it did in May 2025 over US EDA restrictions.

⁶⁶ "Nvidia's Jensen Huang: 'China Is Going to Win the AI Race,' FT Reports," *Reuters*, November 6, 2025, <https://www.reuters.com/world/asia-pacific/nvidias-jensen-huang-says-china-will-win-ai-race-with-us-ft-reports-2025-11-05/>.

⁶⁷ Schneider and Chow, "The Future of Economic Security."

⁶⁸ McGuire, "China's AI Chip Deficit."

⁶⁹ Demetri Sevastopulo, "US Must Suspend Nvidia AI Chip Exports to China, Senators Say," *Financial Times*, March 24, 2026, <https://www.ft.com/content/556e534d-bbc5-46e0-8965-ec3a13a8871a?syn-25a6b1a6=1>.



5. China's response

5.1 Beijing's counter-leverage

Despite Western efforts to constrain China's chip self-sufficiency, China maintains sizable access to global computing power. While the precise amount is difficult to quantify, a bottom-up analysis from the think tank ChinaTalk estimates that the country has access to a computing capacity equivalent to more than one-eighth of the world's total (a median estimate of 2.8mn H100-equivalent GPUs).⁷⁰

Stockpiling and chip smuggling

Even before the US administration partially relaxed export controls on Nvidia's H200 chips, China was able to maintain some access to US advanced chips through various means.⁷¹ Until the end of 2023, China had powerful access to foreign AI chips, with estimates suggesting that legal purchases from foreign providers accounted for over 460,000 H100-equivalent GPUs.⁷² After the full crackdown on Nvidia's chip exports took effect in 2024, reports indicate that China began to acquire chips through unofficial channels, with the Center for a New American Security estimating that approximately 140,000 chips were smuggled into China that year alone, a figure that likely increased in 2025.⁷³ Significant stockpiling represents another strategy: CISI reported that Huawei was able to procure enough HBM chips, primarily via Samsung in South Korea, to support at least a full year's worth of production.⁷⁴

Finally, a substantial portion of China's computing capacity is reportedly accessed remotely via the cloud. According to ChinaTalk estimates, the median of this access is roughly 1,026,000 H100e, though it notes that the variability is high due to weak controls against remote access.⁷⁵ This would mean that the bulk of China's compute comes from its remote access.

These factors demonstrate that, even before the US government's approval of Nvidia H200 shipments in December 2025, China has proven its ability to grow, adapt and innovate using globally available technologies, a resilience underscored by Huawei achieving a record year in revenue in 2025, despite US restrictions.

Materials & market power

China has also leveraged its own structural advantages in the chip supply chain to maintain some concessions. As the world's largest electronics manufacturing hub, China consumed 31% of US chip sales in 2023.⁷⁶ Its dominance in the end-user market has made global chipmaking companies like Qualcomm, Micron, and TSMC reliant on China to buy their chips and assemble them into electronic end goods. With the country representing an estimated 35–40% of global end-demand for chips, the loss of access to this market can wipe out tens of billions of dollars from a chipmaker's valuation.

This dynamic was illustrated following the 2019 export controls on Huawei. As Huawei was forced to curtail its smartphone production, companies that supplied chip components to Huawei were hit by the decline in demand. The public backlash from industry leaders, combined with diplomatic negotiations between Trump and Xi, led the US to place a 90-

⁷⁰ "Compute" here means the number of chips within China plus the number of chips China can remotely access abroad. Aqib Zakaria, "How Much Compute Does China Have?" *Chinatalk*, April 7, 2026, <https://www.chinatalk.media/p/how-many-chips-does-china-have>.

⁷¹ One famous example being a former Micron employee, who stole intellectual property for China's UMC in 2020. See "Taiwan Company Pleads Guilty to Trade Secret Theft in Criminal Case Involving PRC State-Owned Company," US Department of Justice, October 28, 2020, <https://www.justice.gov/usao-ndca/pr/taiwan-company-pleads-guilty-trade-secret-theft-criminal-case-involving-prc-state-owned>.

⁷² Zakaria, "How Much Compute Does China Have?"

⁷³ Erich Grunewald and Tim Fist, "Countering AI Chip Smuggling Has Become a National Security Priority," Center for a New American Security, June 11, 2025, <https://www.cnas.org/publications/reports/countering-ai-chip-smuggling-has-become-a-national-security-priority>.

⁷⁴ McGuire, "China's AI Chip Deficit."

⁷⁵ Zakaria, "How Much Compute Does China Have?"

⁷⁶ "Winning the Chip Race," Semiconductor Industry Association, 2025.



day reprieve on Huawei.⁷⁷ At the same time, while Huawei's international phone sales were impacted by the controls, the company's 5G business was relatively unaffected as it was able to stockpile a year's worth of integrated circuits and other components. The Huawei incident demonstrates the difficulty of isolating a major Chinese firm so deeply integrated into the global supply chain ecosystem.

In another example of how deeply entwined one segment of the supply chain process is with another, Nvidia's "China-compliant" A800/H800 product line exists solely to keep some access to Chinese demand under US rules—an example of how corporate strategies are now being shaped by geopolitics. So, while the US seeks to keep China lagging, at the same time, it cannot leave it too far behind.

US export controls have also galvanized China's pursuit of domestic self-sufficiency. In response to the export controls announced in 2019, the CEO of Huawei's chip development arm, HiSilicon, noted that technologies long developed as "spare tires" had "become the main tires" overnight. This reflects a broader trend: increased US pressure has strengthened China's resolve to vertically integrate its domestic semiconductor industry. This ambition is apparent in the PRC's draft of its 15th five-year plan (2026-2030), which calls for "extraordinary measures" to achieve "full-chain decisive breakthroughs" across EDA software, manufacturing equipment, and advanced materials. The plan also acknowledges that, while China is good at quickly scaling up existing tech, it needs to improve domestic innovation in basic research and discovery. To support this, the government plans on establishing a national M&A fund to improve the exit environment for venture capital.

5.2 China remains compute-constrained

Despite its measurable success in countering certain US chip controls, China still lacks the critical technologies required to shore up its own domestic supply chains. The reason is that the country faces two significant constraints: access to top-tier talent and access to advanced logic compute.

The first constraint is human capital. As our China analysts point out [here](#), while China is the leading source of top-tier AI researchers, a majority of them ultimately end up working in the United States. The recent success of DeepSeek's team, which was overwhelmingly domestically educated, suggests that if Chinese universities and companies can offer conditions and job opportunities comparable to the US for AI talent, the country can better retain its talent pool.⁷⁸ Until then, however, China is likely to continue to experience a net outflow of its top AI researchers to the US.

The more serious barrier is access to advanced logic compute. An analysis from the Council on Foreign Relations (CFR) finds that the leading US AI chips are approximately 5x more powerful than China's current domestic alternatives from firms like Huawei.⁷⁹ The same analysis projects that this performance gap will continue to grow and could widen to as much as 17x.

Unless China can produce its own advanced chips, it will continue to rely on Taiwan for manufacturing. And it cannot build its own advanced chip supply without critical EUV lithography tools, a market monopolized by the Dutch firm ASML. In alignment with US policy, ASML has restricted the sale of its most advanced equipment to Chinese chipmakers, preventing them from building fabs capable of producing leading-edge chips and reducing their reliance on Taiwan. As a result, Chinese manufacturers are effectively stalled at the 7nm node.

⁷⁷ For more on sanctions and chokepoints, read Edward Fishman.

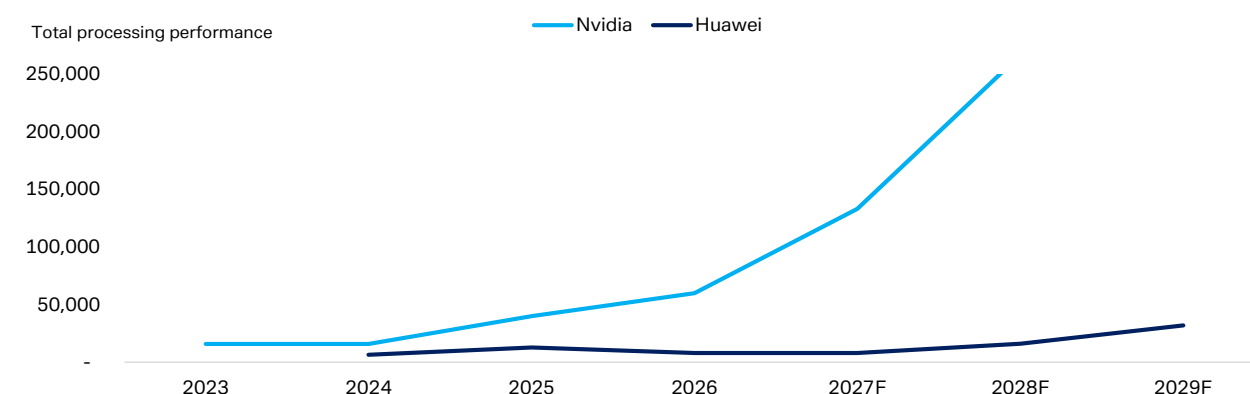
⁷⁸ Yi Xiong, "China Macro: The Upcoming AI Boom in China," Deutsche Bank Research.

⁷⁹ McGuire, "China's AI Chip Deficit"



While Chinese companies like Xiaomi and Huawei are designing and developing their own semiconductors, the process is very costly and expensive. Although SMIC is reportedly moving toward mass production of 7nm and potentially 5nm advanced chips within the next one to two years, significant doubts persist about its ability to achieve the necessary scale and yield.⁸⁰ Even if successful, the technological frontier continues to advance, with the US and Taiwan already producing at the 2nm node, and some companies reportedly already planning for 1.4nm technologies. The cumulative result of these constraints is that China's frontier AI models have lagged their US counterparts by an estimated seven months since 2023.⁸¹

Figure 9: Nvidia vs Huawei's AI chips – total processing performance (TPP) comparison



Note: Based on the best chip available in H1 of each year.

Source: Deutsche Bank, Council on Foreign Relations, Nvidia Roadmap, Huawei Roadmap, SemiAnalysis.⁸²

5.3 China is establishing a competitive advantage in a US blind spot

While China lags the leading-edge competition, it is rapidly establishing global dominance in legacy chips. This development has been enabled, in part, by a US policy framework that has not cohesively addressed either domestic legacy chip production or China's growing strength in this segment. Thus, US export controls have left China's access to legacy chip technology largely unaffected, and China's overall chip imports remain in line with pre-pandemic trends.⁸³ This has led to warnings from bodies like the US National Security Council that, if unaddressed, the US could become dependent on China for components used in even the most "mundane" electronic devices, creating a new point of strategic leverage for Beijing.⁸⁴

In fact, Beijing has made massive investments in recent years to build China's dominance in mature-node chips, utilizing strategies ranging from accelerated fab production to import-substitution policies that guarantee domestic demand for its chip companies.

The US China Security Council predicts that China's share of global production could rise to 47% by 2027 from 34% in 2024.⁸⁵ In contrast, Taiwan's share is expected to decline from 43% to 36% over the same period. Should China establish hegemony over foundational chips, it could compel US manufacturers to rely on Chinese suppliers for essential components as non-Chinese competitors are displaced from the market. Non-Chinese entities, like Global Foundries (for chip manufacturing) and Texas Instruments (for integrated devices) are increasing their share in the semiconductor supply chain to reduce this risk.

⁸⁰ "China Reportedly Aims to Boost 7nm, 5nm Output Fivefold in Two Years, Driven by SMIC, Hua Hong," TrendForce, February 25, 2026, <https://www.trendforce.com/news/2026/02/25/news-china-reportedly-aims-to-boost-7nm-5nm-output-fivefold-in-two-years-driven-by-smic-and-hua-hong/>.

⁸¹ Luke Emberson, "Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023," Epoch AI, 2023, <https://epoch.ai/data-insights/us-vs-china-ecil/>.

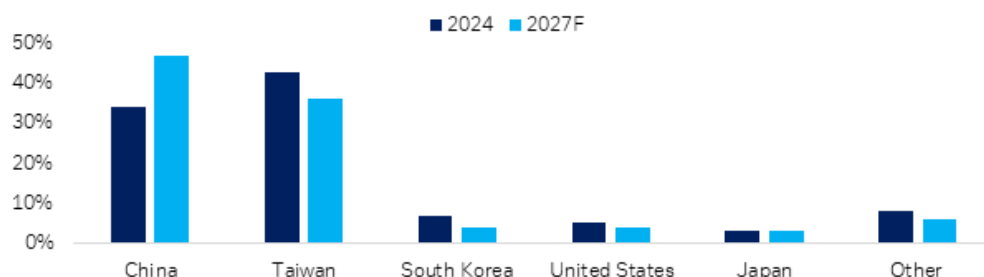
⁸² McGuire, "China's AI Chip Deficit."

⁸³ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

⁸⁴ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

⁸⁵ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

Figure 10: Foundational foundry capacity by location, 2024 and 2027 (forecast)



Source: Based on USCC analysis of TrendForce Data, Deutsche Bank Research.

Finally, the global focus on high-bandwidth memory (HBM) has also created an opportunity for China's semiconductor exports. According to Chinese customs data, memory chips accounted for 52.8% of the country's total semiconductor exports in December 2025, as electronics companies have turned to Chinese suppliers amidst the memory supply glut.⁸⁶ US opposition aside to Chinese firms (e.g. CXMT and Yangtze Memory Technologies Corp) Western companies rely on China's expansion in this segment to support conventional memory, not HBM.

If this trend persists, China might end up being a winner in this market. If memory chip supply is unable to meet demand, average selling prices could continue to rise through 2026, potentially benefiting Chinese producers. However, some analysts believe it is still too early to tell if China can sufficiently fill these supply gaps.⁸⁷ This is why algorithmic developments like Google's TurboQuant could have major implications for China; if efficiency gains stemming from the algorithm reduce the physical memory capacity required by data centers, future demand may not be as high as currently projected.⁸⁸ Nevertheless, multiple industry sources suggest that a surplus in the market for HBM seems unlikely in the near term.

6. Global implications & industry response

6.1 Impact on industry

The trade wars and retaliatory strategies between China and the US in 2025 created arguably peak volatility for the semiconductor market. This was followed by a V-shaped recovery in chip equities (post a bottoming in April 2025), as investors were reassured by company-specific mitigation strategies and global renegotiation. The Philadelphia Stock Exchange Semiconductor Index (SOX) recovered to end 2025 up 41% YTD – roughly 25pp ahead of the S&P 500 and more than 20pp ahead of the Nasdaq 100 in 2025.⁸⁹ Even with the geopolitical tensions of 2025, seasonal semiconductor sales were not meaningfully impacted, suggesting continued fundamental strength in chip demand.⁹⁰

⁸⁶ General Administration of Customs, People's Republic of China.

⁸⁷ Irene Zhang and Nick Corvino, "Review of Chinese AI Rings in the Year of the Horse," *ChinaTalk* (blog), February 18, 2026, https://www.chinatalk.media/p/chinese-ai-rings-in-the-year-of-the?utm_campaign=email-half-post&r=6q6sor&utm_source=substack&utm_medium=email.

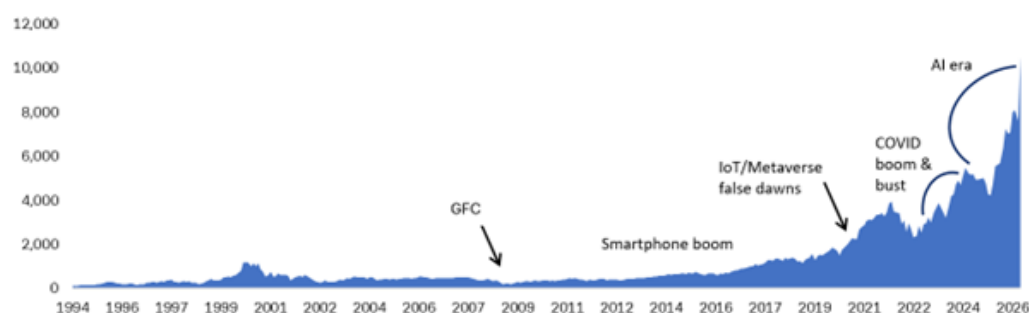
⁸⁸ Tim Bradshaw and George Steer, "Memory Chip Stocks Shed \$100bn as AI-Driven Shortage Trade Unwinds," *Financial Times*, March 27, 2026, <https://www.ft.com/content/e4e15692-187e-4466-832e-ec267e792292?syn-25a6b1a6=1>.

⁸⁹ Bloomberg Finance LP.

⁹⁰ Ross Seymore, Apoorva Kumar, DJ Sebastian, and Melissa Weathers, "Semiconductors: 2026 Outlook: History Doesn't Repeat, but It Rhymes (Again)," Deutsche Bank Research.



Figure 11: Semiconductor index (SOX)



Source: Bloomberg Finance LP, Deutsche Bank

6.2 The AI race and its effect on the chip industry

As AI infrastructure investment continues to be driven by Chinese and US companies, we believe the industry should expect to see a continued bifurcation between AI leaders and broad-based laggards. As our equity analysts point out in their note [here](#), investor focus on AI is broadening beyond GPUs to related segments such as ASICs, networking, memory, and power management solutions.

Looking to 2026 and beyond, the rate of hyperscaler capex will continue to impact chip prices. While key hyperscalers maintain their air of confidence and continue to announce AI-related spending deals (Sovereign AI, OpenAI, Anthropic, etc.), many of these deals are not yet set in stone and are more akin to agreements or MOUs rather than finalized contracts. Investors are increasingly worried about the return on investment on these massive investments, with concerns about the "monetization" of AI beginning to overtake the focus on continued acceleration. For example, Microsoft's inability to meet growth expectations for its Azure cloud division – attributed to a lack of advanced chips – was the precipitating factor in the company's \$357bn single-day market value loss earlier this year. Still, Q1 2026 earnings from Big Tech revealed more spending by large cloud providers and strong cloud / AI income than anticipated, suggesting the AI trend will drive significant growth for chip companies. Our equity analysts expect the total chip market to reach \$969bn (+25% y/y) in 2026, driven by capital expenditures from these large cloud providers.⁹¹

Looking ahead, market participants should continue to brace themselves for continuous AI-related disruption. The most recent example is Google's TurboQuant algorithm announcement, which suggests that companies may require fewer GPUs/TPUs to serve large language models (LLMs). The announcement caused a sharp decline in the share price of memory providers like Samsung (-23%), Micron (-18%) and SK Hynix (-24%) in March.⁹² However, whether the TurboQuant technique will create a structural shift in demand remains to be seen. A key indicator to watch is actual order behaviour; unless top suppliers report significant HBM order cancellations, it is difficult for us to envision a substantial reduction in memory demand.

6.3 Impact on memory chip demand

Notwithstanding disruptions like the TurboQuant announcement, the risk of memory chip shortage in everyday standard DRAM and NAND – the integrated circuits used for phones, cars, computers – is becoming more acute. This concern has arisen as memory giants like Samsung turn their attention and resources toward developing HBM for AI instead, which as we have said, requires vast amounts of high-performance memory. For example, Micron, a key memory producer, is described as "supply-constrained" and only able to meet "50-66% of customer demand in the medium term" despite efforts to increase

⁹¹ Seymore et al., "Semiconductors: 2026 Outlook."

⁹² Bloomberg Finance LP.



supply.⁹³ The broader memory market is now expected to remain "supply-constrained through 2027".⁹⁴

As a consequence, companies in consumer-facing markets are warning of potential slowdowns in PC and handset sales due to the rising cost of memory, which makes the final products more expensive. Qualcomm, for instance, is seeing "significant headwinds" in its Handsets segment due to elevated memory pricing.⁹⁵ Other industry leaders now describe memory as a more critical constraint than even semiconductor manufacturing capacity, with the CEO of ARM noting in a recent interview that he is now less concerned about TSMC fab allocation than he is about memory.⁹⁶

6.4 The role of Europe, Japan, South Korea

Europe, South Korea and Japan play critical roles in the semiconductor supply chain but find themselves caught in the middle of the complex strategic competition between the US, Taiwan and China.

For Europe, the EU's Chips Act has been in force since September 2023 with limited initial results. The legislation, introduced in response to the supply chain disruptions caused by the COVID pandemic, set a goal for the EU to achieve a 20% share of global chip production by 2030. However, a recent audit by the European Court of Auditors concluded that the initiative is unlikely to meet its target in time, and is insufficient to stimulate the level of investment needed for Europe's chips industry.⁹⁷ As a result, the EU's overall share of the global value chain is projected to reach only 12% by 2030, despite its significant increase in manufacturing capacity.

ASML, a Dutch chip equipment maker, is Europe's key player in this industry. The company plans to produce at least 60 EUV machines this year and at least 80 in 2027.⁹⁸ However, some critics argue that Europe is not leveraging this asset effectively. The Dutch government's decision to align with US policy and restrict exports of both EUV and DUV (deep ultraviolet) machines to China has impacted ASML's earnings and demonstrated Europe's dependence on US strategic direction, turning Europe's strongest card into just another constrained asset.

In South Korea, the country's primary advantage has been its leadership in high-end memory. Until late 2024, when US policy changes required Seoul to halt the manufacturing of 7nm and more advanced chips for Chinese AI firms, a considerable portion of Chinese-designed AI chips were manufactured by Samsung. Since then, the South Korean government has pursued a dual strategy: while aiming to closely align itself with the US policies around semiconductors, it has also increasingly focused on boosting its domestic material sourcing and expanding into the logic and AI chip markets. These efforts are backed by significant government support, including a March 2023 law providing tax credits of 15-35% for semiconductor investments. The country's aim is to locally source 50% of its semiconductor materials by 2030, up from the current 30%.

Meanwhile, Japan has also played a key role in the effort to manage China's technological advancement while bolstering its own industry. Under Sanae Takaichi, Japan's government has committed to boosting its domestic semiconductor industry to reach \$254bn in annual sales by 2040. The country's flagship project, Rapidus, has received over \$11 bn in

⁹³ Joseph Wilkins, "Micron CEO Says It Can't Deliver Enough Memory to Key Customers after Blowout Earnings," CNBC, March 19, 2026, <https://www.cnbc.com/2026/03/19/micron-falls-blockbuster-earnings-chips-ai.html>.

⁹⁴ Ross Seymore, Apoorva Kumar, and Melissa Weathers, "Global Memory: Why this cycle is different; Updating Demand-Supply model; MU T/P to \$500," Deutsche Bank Research, Feb 10 2026.

⁹⁵ Ross Seymore, DJ Sebastian and Melissa Weathers, Feb 5 2026. "Qualcomm: F1Q26 (Dec) Results: Memory loss," Deutsche Bank Research.

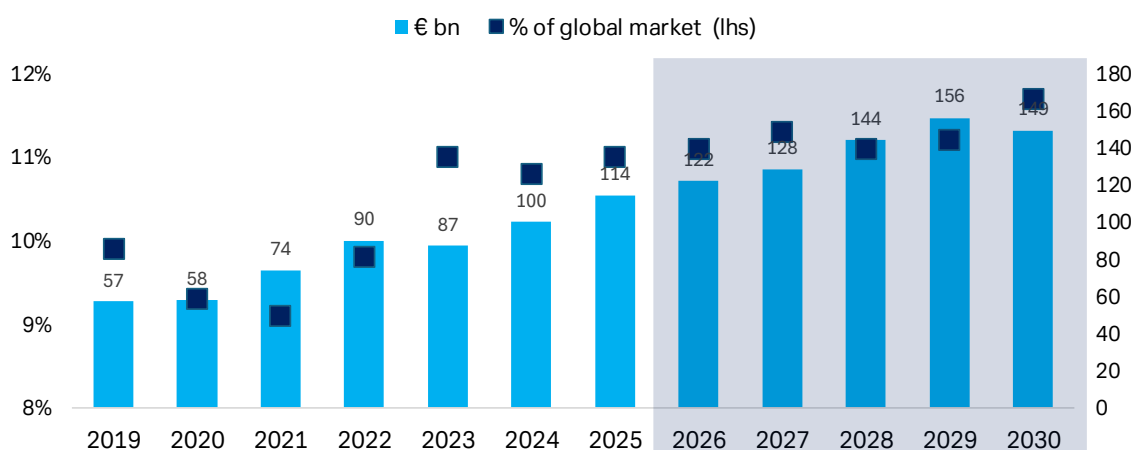
⁹⁶ Ben Thompson, "An Interview with Arm CEO Rene Haas about Selling Chips," Stratechery (podcast), 2026, <https://stratechery.com/2026/an-interview-with-arm-ceo-rene-haas-about-selling-ch>.

⁹⁷ "Special Report 12/2025: The EU's Strategy for Microchips – Reasonable Progress in Its Implementation but the Chips Act Is Very Unlikely to Be Sufficient to Reach the Overly Ambitious Digital Decade Target," (Luxembourg: European Court of Auditors, 2025).

⁹⁸ Mackrael, Kim. 2026. "The Race to Make the World's Most In-Demand Machine." *The Wall Street Journal*. April 25, 2026. <https://www.wsj.com/business/the-race-to-make-the-worlds-most-in-demand-machine-092e8cea>.

state investment and aims to reach 2nm production (currently from 40nm) by the late 2020s. The latest technical milestone was reached in July 2025, when Rapidus announced it had successfully prototyped a 2nm-generation GAA transistor at its IIM-1 manufacturing facility in Chitose, Hokkaido. And in another win for Japan this year, TSMC unveiled plans to produce 3nm chips in Japan, following heavy negotiations with the Japanese government to secure funding for its expansion into the country.⁹⁹

Figure 12: 2030 forecast of EU-27 share of global market value chain



Source: European Court of Auditors, COM (2024), based on the International Data Corporation's study in 2022.

7. Conclusion: Silicon sovereignty unlikely in the near to mid-term

The U.S.-China Economic and Security Review Commission has framed critical minerals as the “immediate” challenge in the bilateral relationship between the US and China, whilst viewing semiconductors as a “future chokepoint.”¹⁰⁰ This suggests that in the short-term, China can exert leverage at the bargaining table because its control over critical minerals represents a more immediate global bottleneck, whereas the semiconductor dynamic has a longer time horizon.

While this assessment has merit, this chapter has sought to demonstrate that the semiconductor supply chain is already filled with chokepoints, making chips a latent but escalating threat. The US will likely need 15 more years to develop the necessary capacity and skills to establish a chip manufacturing and packaging industry that does not rely on Taiwan, not to mention the additional costs involved.

The result is a state of complex interdependence: while China continues to struggle to produce chips on par with those from the US, Europe, Japan, South Korea and Taiwan, these allied nations also face significant hurdles in achieving full industrial sovereignty and breaking their reliance on one another. At the same time, China is growing increasingly more competitive in areas where US policy is less focused, such as legacy and foundational memory chips.

This complex reality helps explain why the current US administration's partial relaxation of advanced AI chip controls has split stakeholders and perplexed some analysts. At Davos in January, Anthropic's CEO likened the sale of advanced US chips to China to “selling nuclear weapons to North Korea.”¹⁰¹ In contrast, others argue that because the US relies on TSMC in Taiwan for its most advanced AI chips, a complete technological cutoff of

⁹⁹ Harry Dempsey and Kathrin Hille, “TSMC to Produce Advanced Chips in Japan,” *Financial Times*, February 5, 2026, <https://www.ft.com/content/83e0fd2a-1fa4-4004-9ffb-ef929075a497?syn-25a6b1a6=1>.

¹⁰⁰ “Chapter 9: Chained to China,” U.S.-China Economic and Security Review Commission.

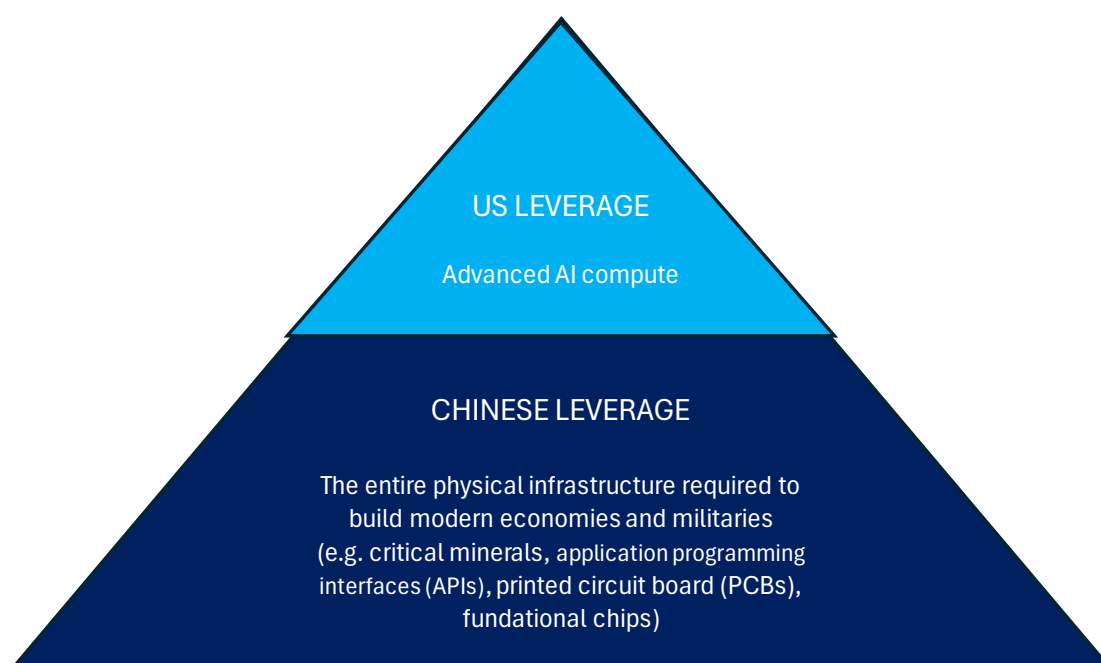
¹⁰¹ Christina Kyriasoglou, “Anthropic CEO Says AI Chip Sales to China like Selling Nukes to North Korea,” *Bloomberg.com*, January 20, 2026, <https://www.bloomberg.com/news/articles/2026-01-20/anthropic-ceo-says-selling-advanced-ai-chips-to-china-is-crazy>.



China would incentivize Beijing to adopt a “scorched-earth” strategy towards Taiwan’s foundries.¹⁰² This has led some experts to advocate for a policy of keeping China dependent on the US-based ecosystem, while simultaneously denying it the advanced EUV equipment needed to ultimately create its own leading-edge fabs.

Looking ahead to the next few chapters, one of the wider themes emerging from this series is that the chip competition is increasingly about the race for AI dominance. The outcome of this race may depend on its timeline: a rapid evolution of AI likely favors the US due to its massive capital and compute advantages, whereas a slower, more industrial-focused timeline could favour China due to its superior physical supply chains and manufacturing scale. In other words, China could arguably win from an industrialization standpoint, while the US can lead by diffusing its AI capabilities. But the limitations of US industrial capacity, as highlighted by the mixed results of the CHIPS Act and its latest export controls, have created an opening for private corporations to tackle the production problem.¹⁰³ The influence of firms like Nvidia, for example, was a significant factor in the administration’s decision regarding H200 chip sales to China. Thus, in this era of “infrastructure realism” driven by the pursuit of AI supremacy, perhaps the more accurate way of surveying the strategic actors will be via companies rather than countries or governments. We will explore this theme further in our forthcoming chapter on cloud computing.

Figure 13: The US maintains a lead in advanced AI, but China holds the industrial capacity



Source: Deutsche Bank Research. An API is a set of defined methods of communication among various software components. In the context of semiconductors or design, it could refer to interfaces for software tools or IP blocks. A PCB is a board used to mechanically support and electrically connect electronic components using conductive pathways, tracks, or signal traces etched from copper sheets laminated onto a non-conductive substrate. They are fundamental in nearly all electronic devices.

¹⁰² Leading tech analyst Ben Thompson instead advocates for keeping China dependent on a US-based ecosystem.

¹⁰³ Tesla CEO Elon Musk announced the construction of a massive factory called “TerraFab” capable of producing AI chips at an annual scale of 1 terawatt (TW). This decision stems from the assessment that current AI chip production volumes are insufficient to meet demand. The project will involve all three of Musk’s companies: Tesla, SpaceX, and xAI.



Appendix 1

This material has been prepared by the Deutsche Bank Research Institute and is provided to you for general information purposes only. The Institute leverages the views, opinions, and research from Deutsche Bank Research, economists, strategists, and research analysts. Accordingly, you should assume that content in this document is based on or was previously published and provided to Deutsche Bank clients who may have already traded on the basis of it.

Any views or estimates expressed in this material reflect the current views of the author(s) and may differ from the views and estimates of Deutsche Bank AG, its affiliates, other Deutsche Bank personnel, and other materials published by Deutsche Bank. The content in this material is valid as of the date shown on the first page and may change without notice. Deutsche Bank has no obligation to provide any updates or changes to the information herein.

This material should not be used as a basis for trading securities or other financial products and should not be considered to be a recommendation or individual investment advice for any particular person. It does not constitute an offer, solicitation, or an invitation by or on behalf of Deutsche Bank to any person to buy or sell any security or financial instrument. Nothing in this material constitutes investment, legal, accounting or tax advice. Deutsche Bank engages in securities transactions, including on a proprietary basis, and may do so in a manner inconsistent with the views or information expressed in this material.

While information in this material has been obtained from sources believed to be reliable, neither Deutsche Bank AG nor any of its affiliates makes any representation or warranty, express or implied, as to the accuracy or completeness of the statements or any information contained in this material and therefore any liability is expressly disclaimed. This material is provided without any obligation, whether contractual or otherwise. Information regarding past transactions or performance is not indicative of future results.

In the U.S. this report is approved and/or distributed by Deutsche Bank Securities Inc., a member of FINRA. In Germany this information is approved and/or communicated by Deutsche Bank AG Frankfurt, licensed to carry on banking business and to provide financial services under the supervision of the European Central Bank (ECB) and the German Federal Financial Supervisory Authority (BaFin). In the United Kingdom this information is approved and/or communicated by Deutsche Bank AG, London Branch, a member of the London Stock Exchange, authorized by UK's Prudential Regulation Authority (PRA) and subject to limited regulation by the UK's Financial Conduct Authority (FCA) (under number 150018) and by the PRA. This information is distributed in Hong Kong by Deutsche Bank AG, Hong Kong Branch, in Korea by Deutsche Securities Korea Co. and in Singapore by Deutsche Bank AG, Singapore Branch. In Japan this information is approved and/or distributed by Deutsche Securities Limited, Tokyo Branch. In Australia, retail clients should obtain a copy of a Product Disclosure Statement (PDS) relating to any financial product referred to in this report and consider the PDS before making any decision about whether to acquire the product.

By accessing this material, you agree that its content may not be reproduced, distributed or published by any person for any purpose, in whole or part, without Deutsche Bank's prior written consent. You also agree that you shall not scrape, extract, download, upload, sell or offer for sale any of the content in this material, and you agree not to use, or cause to be used, any computerized or other manual or automated program or mechanism, tool, or process, including any scraper or spider robot, to access, extract, download, scrape, data mine, display, transmit, or publish, any of the content in this material.